

BUNDLED PRICING AND METERED PRICING

Overview

2

Three pricing structures

1 **Bundling** to capture value

2 **Mixed Bundling** as product line pricing

3 **Metered Pricing**

- ▣ The “Razors & blades problem”
- ▣ Related to bundling complements

3 Bundling & Mixed Bundling

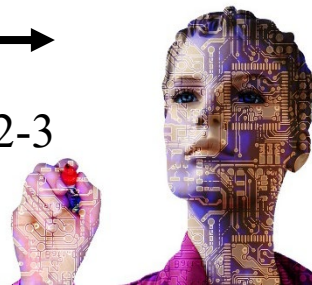
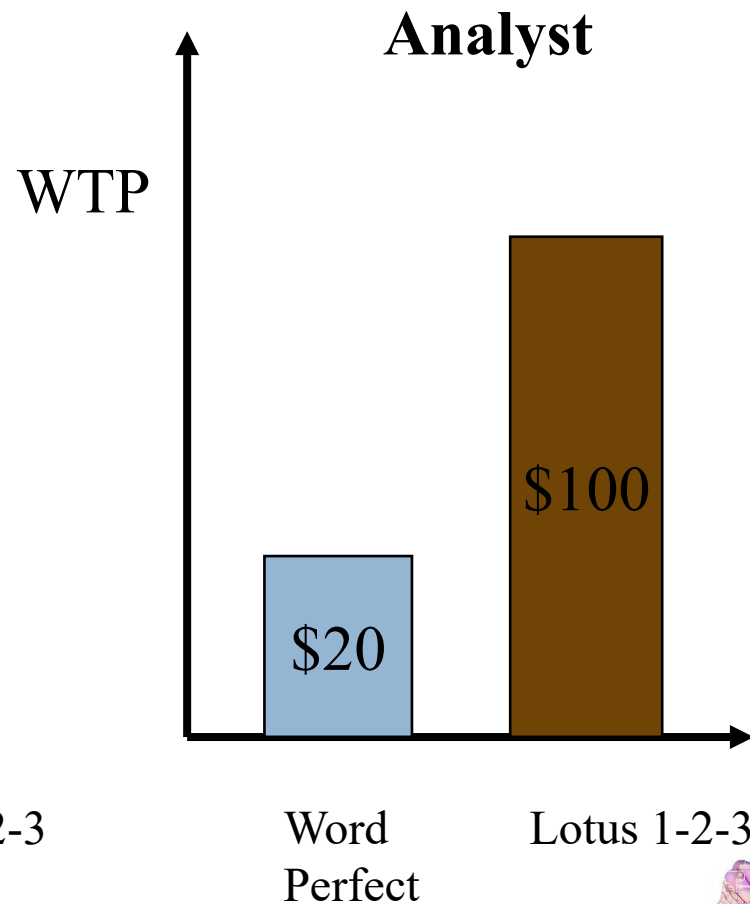


Price Bundling

Bundling non-complements

ni the 80s, 2 main office softwares. Dominant is Word Perfect.
Spreadsheets are Lotus 123.

4

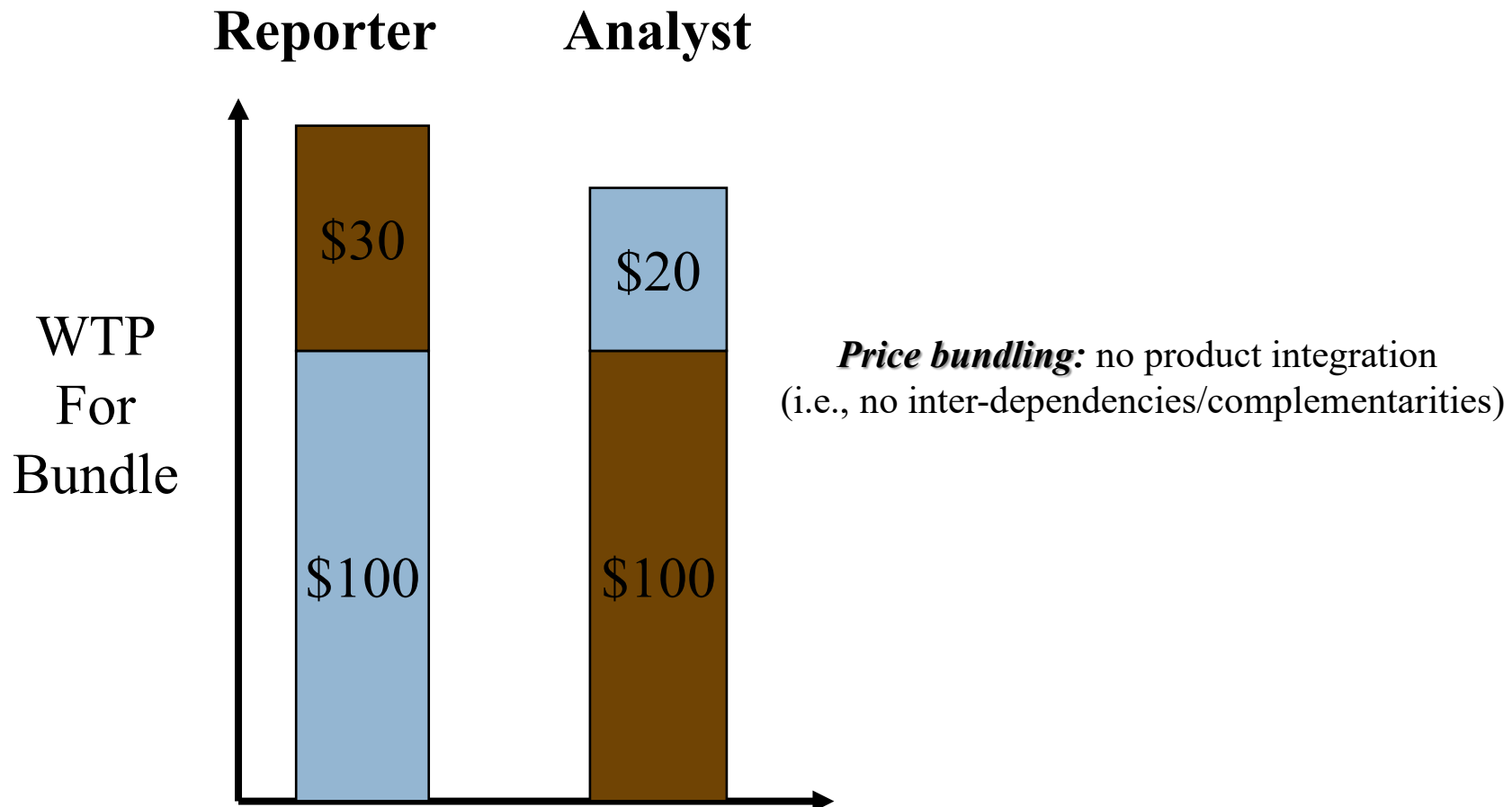


Price Bundling

Bundling non-complements

5

Suppose a monopolist sold both software titles as a **bundle**?



Price Bundling

Bundling non-complements

6

□ Pure Components

- *Word Perfect*: \$100, give up Analysts
- *Lotus 1-2-3*: \$100, give up Reporters
- Get \$100 total from each customer
 - Untapped demand: \$30 for Reporters, \$20 for Analysts

□ Pure Bundling

- Bundle: \$120, sell both items to all customers
- Only \$10 left on table for reporters
- **i.e., Bundling to price-discriminate (*capture more value*)**

Price Bundling

Bundling non-complements

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Why?

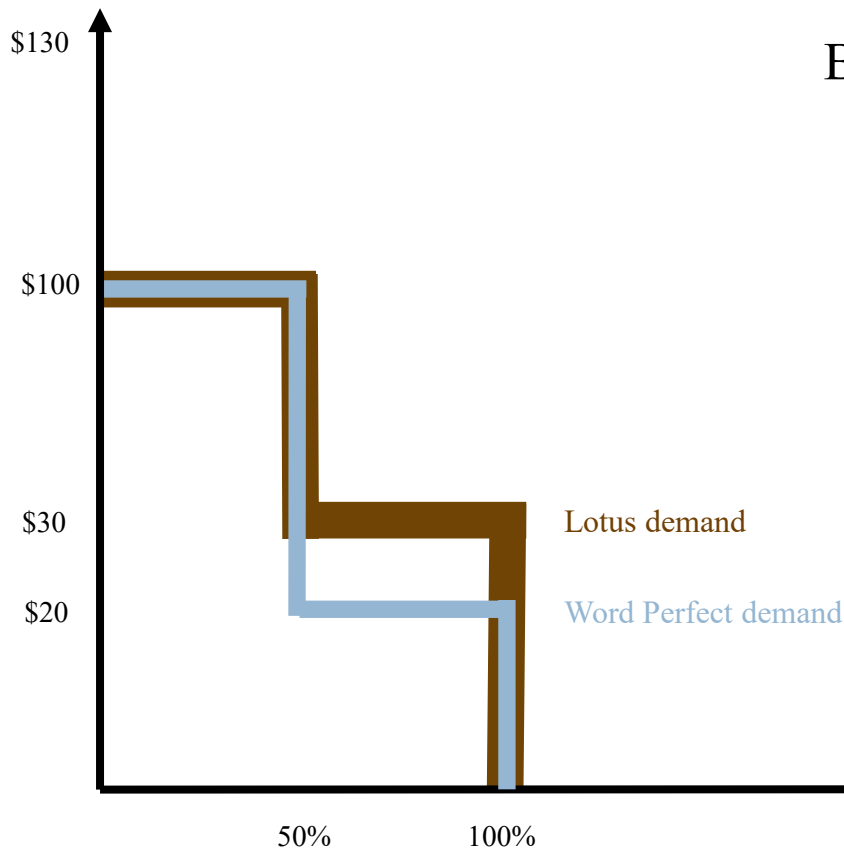
Price Bundling

Bundling non-complements

7

Compare Market Demands

Price



Result:

Bundle demand is more elastic
(less dispersion in WTP)

Price Bundling

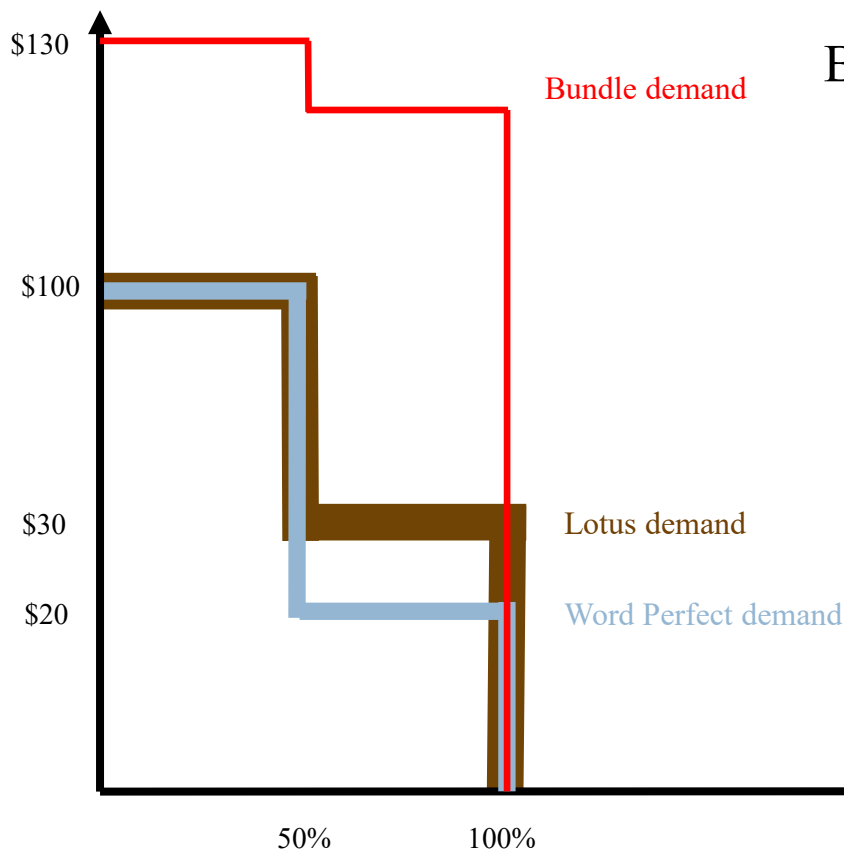
Bundling non-complements

7

Compare Market Demands

originally, need to cut price by 70/80% to get 100% more demand.
bundle, 'cut' price by less than 10% to get 100% more demand. Thus, bundle creates a more elastic demand. (also, very little consumer surplus, and very little DWL).

Price



Result:

Bundle demand is more elastic
(less dispersion in WTP)

Price Bundling

Bundling non-complements

8

bundling; implicit price discrim method

- **Bundling** can capture consumer value (i.e., price discriminate) if WTP for bundle more *homogeneous* than WTP for components

- b/c tastes are negatively correlated

➡ bundled demand is more price-elastic

Price Bundling

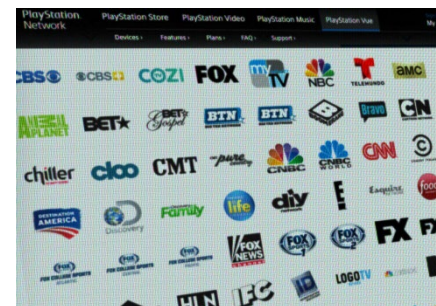
Bundling non-complements

8

- **Bundling** can capture consumer value (i.e., price discriminate) if WTP for bundle more *homogeneous* than WTP for components
 - b/c tastes are negatively correlated
 - ➡ bundled demand is more price-elastic

- **Cable TV** is canonical example for bundling
 - MTV & SoapNet taste correlation: -0.13
 - CNBC & Comedy Central taste correlation: -0.19versus
 - ESPN & ESPN2 have taste correlation of 0.67

Source: [Crawford and Yurukoglu \(2011\)](#)



Price Bundling: competitive strategy

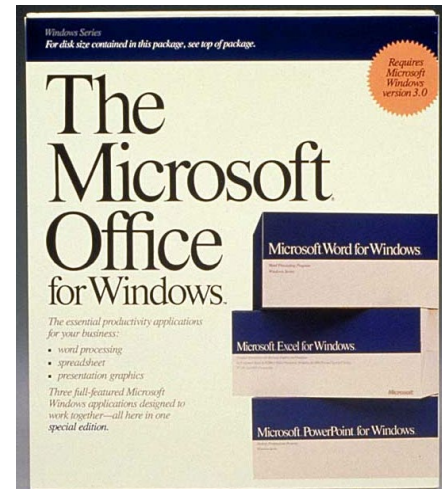
Microsoft's launch of *MS Office* in 1990

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- Office Bundle more appealing
 - ▣ Analysts indifferent
 - ▣ Reporters strictly prefer bundle

- Microsoft's bundle more profitable than incumbents' combined profits.
 - ▣ In example, bundle profits=\$240 vs component profits=\$200.

- See appendix for actual data!



Bundling vs Unbundling

10

□ Free Shipping as *bundling*

Are Shipping Costs Hurting
Your Online Sales?



Cashdrop Brand Contributor
Forbes EQ BRANDVOICE | Paid Program



May 3, 2021, 11:18am EDT



➡ *Unbundle?*

Bundling vs Unbundling

10

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Are Shipping Costs Hurting Your Online Sales?



Cashdrop Brand Contributor
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May 3, 2021, 11:18am EDT



➡ *Unbundle?*

□ Baggage fees as *unbundling*

Bag Fees Are Getting More Costly, Complex

BY ALISON SIDER AND ALLISON POHLE *Wall Street Journal, April 23, 2024*



Bundling vs Unbundling

10

□ Free Shipping as *bundling*

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May 3, 2021, 11:18am EDT

Anytime you hear that you are getting something free, you are actually buying a bundle!



➡ *Unbundle?*

□ Baggage fees as *unbundling*

Bag Fees Are Getting More Costly, Complex

BY ALISON SIDER AND ALLISON POHLE *Wall Street Journal, April 23, 2024*



□ **Partitioning** prices effective if consumers don't pay attention to add-ons at point of sale (see [Greenleaf et al 2016](#))

➔ But is this marketing for bad?

Unbundling, Partitioned Prices & Salience

Consumer **Inattention** (marketing for good?)

11

- Consumers don't consider/notice the “add-on” prices...

- ▣ Demand falls when combine all fees into price



- Sales tax inclusive prices in retail (Chetty et al 2009)

StubHub

- Fee-inclusive prices at stubhub.com (Blake et al 2020)

Unbundling, Partitioned Prices & Salience

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The Gap Between the Price You See and What You Pay Is Getting Worse

A mass unbundling of charges creates more surprise fees when it's time to pay

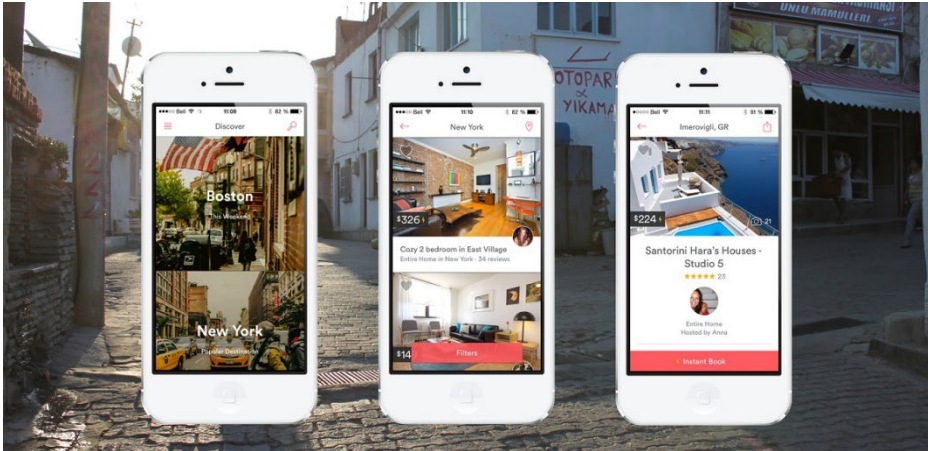
Rachel Wolfe, *WSJ*, May 7, 2024

Legality of hidden fees

Airbnb in Australia & Europe



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Results pages exclude charges & fees from displayed prices

- Australia ([Australian Competition & Consumer Commission, 2015](#)) and Europe ([European Commission, 2019](#)):

“In accommodation searches with selected dates, users see the **total price** in the results page, including **all the applicable mandatory charges and fees** (such as service, cleaning charges and local taxes). There are now no surprise mandatory fees appearing on later pages.”

Legality of hidden fees

Hidden Ticket Fees

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New York to Ban Hidden Fees in Live-Event Ticketing

Anne Steele, WSJ, 6-7-2022

“It is seen as a rare win for consumers who, lawmakers say, have suffered from ticketing companies’ predatory pricing practices.”

- Mandates:
 - ▣ All-in pricing (final price disclosed from start)
 - ▣ Re-sales require displaying original price

Sustainability of Junk Fees

Hotel Resort Fees



14

“Marriott and Hilton Sued Over ‘Resort Fees,’ Long a Bane for Travelers”

8-23-2019, *New York Times*

- Resort fees typically not disclosed with published room rates
- \$9-\$95 per day at Marriott, \$15-\$45 per day at Hilton
- Disclosure only after beginning *checkout* process
- According to [Consumer Reports \(2023\)](#) :
 - ▣ 1/3rd of US Adults charged “hidden” hotel fees in past 3 years
 - ▣ Hotel industry collected \$2.9 Billion in resort fees in 2018
- Bi-partisan [bill H.R. 4489](#) to prohibit such deceptive price advertising

Sustainability of Junk Fees

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AG: Marriott Hotels to pay \$225K
in ‘hidden fee’ settlement
Vivian Muniz, Crime & Court, April 12, 2023

Pending legislation in the U.S.

15

Biden's Open War on Hidden Fees

Targeting “junk fees” from the likes of airlines and credit-card companies has been a way for the president to show Americans that he’s acting to lower prices.

May 22, 2024

The New York Times

Hosted by Michael Barbaro Featuring Jim Tankersley

Produced by Shannon M. Lin, Eric Krupke, Carlos Prieto and Rikki Novetsky

Edited by Lexie Diao Original music by Marion Lozano and Elisheba Ittoop

Engineered by Chris Wood

Mixed Bundling

16

- Sell Bundle along with the Individual Components
- Analogous to Product Line Pricing
 - ▣ E.g., Office Bundle vs. Word, Excel, etc. à la carte
- Similar Trade-offs
 - ▣ Add pure components to acquire new customers
 - ▣ But adding pure components may cannibalize bundle

Mixed Bundling and non-complements

17

Word

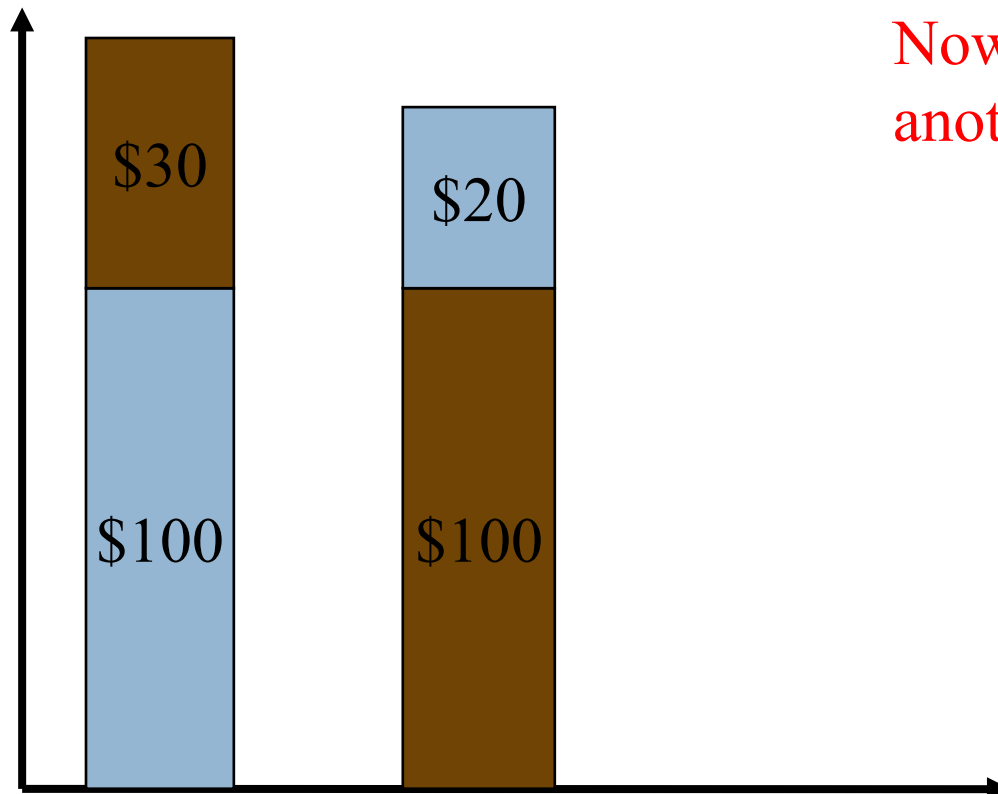
Excel

Reporter **Analyst**

This is our original scenario

Now suppose we have
another segment

WTP
For
Bundle



Mixed Bundling and non-complements

17

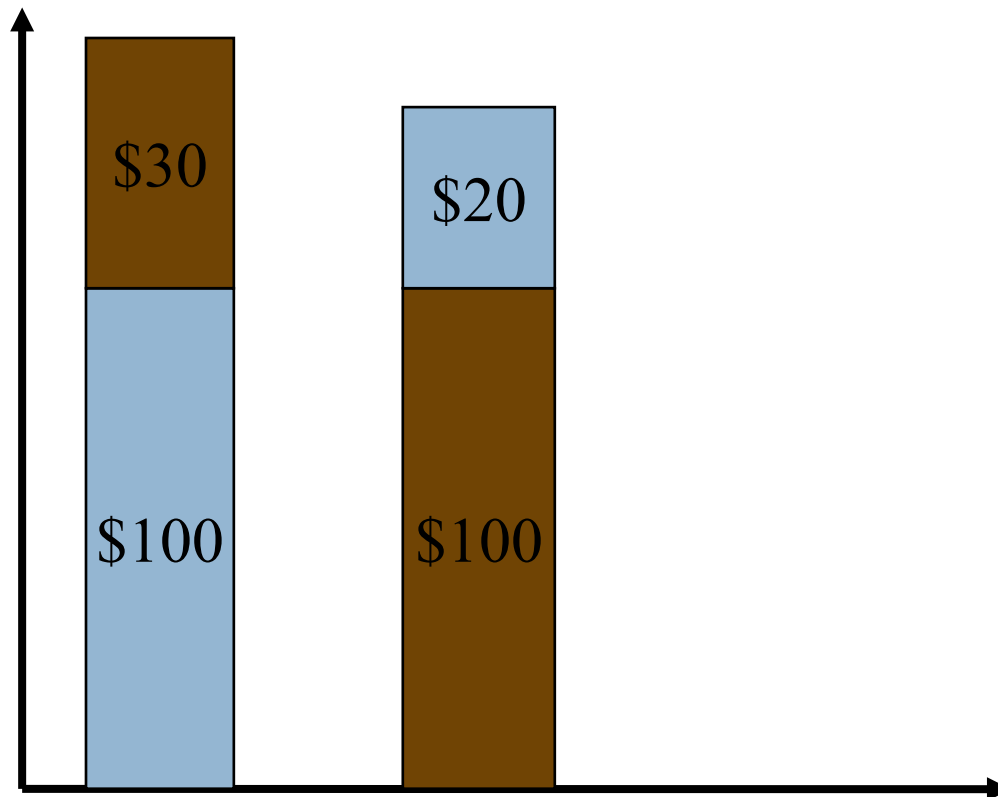
Word

Excel

Reporter

Analyst

WTP
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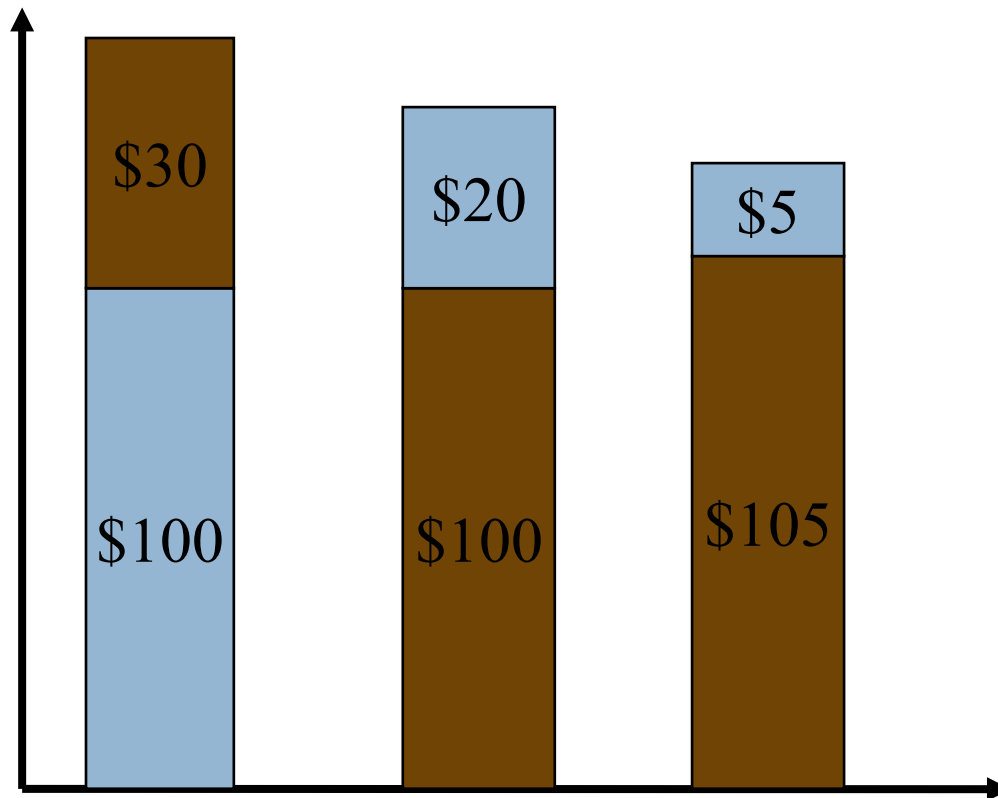
17

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Mixed Bundling and non-complements

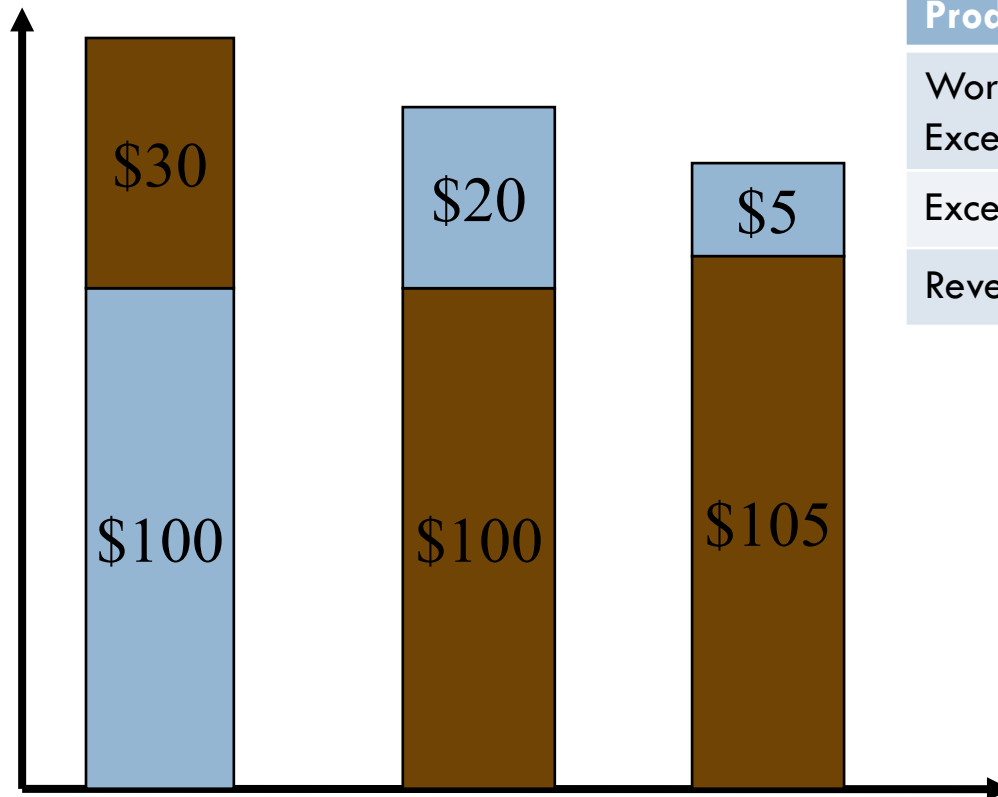
17

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter + Analyst
Excel Only	\$105	Engineer
Revenue	\$345	

Mixed bundling and non-complements

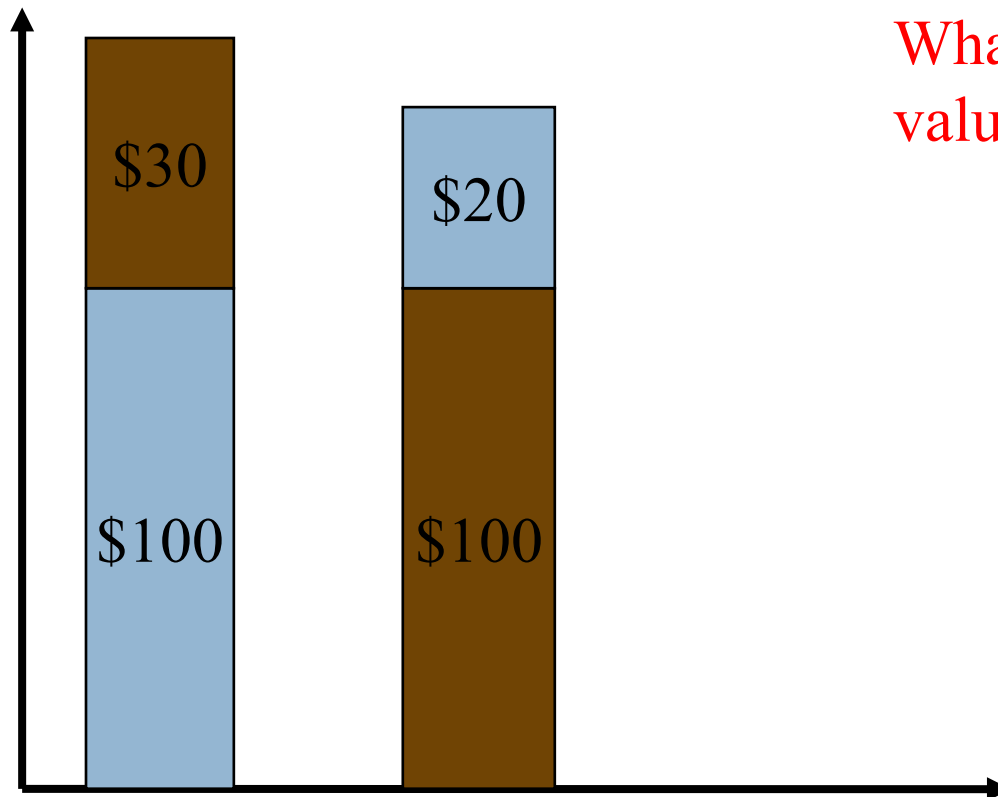
18

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



What if the engineer
valued Excel less?

Mixed bundling and non-complements

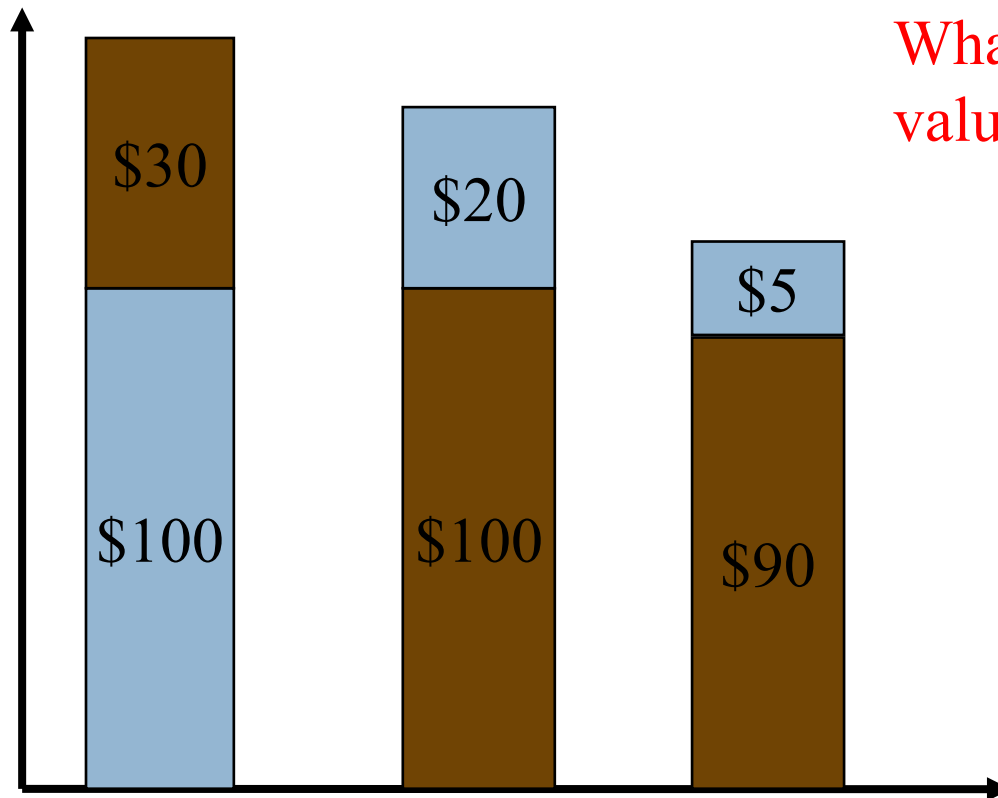
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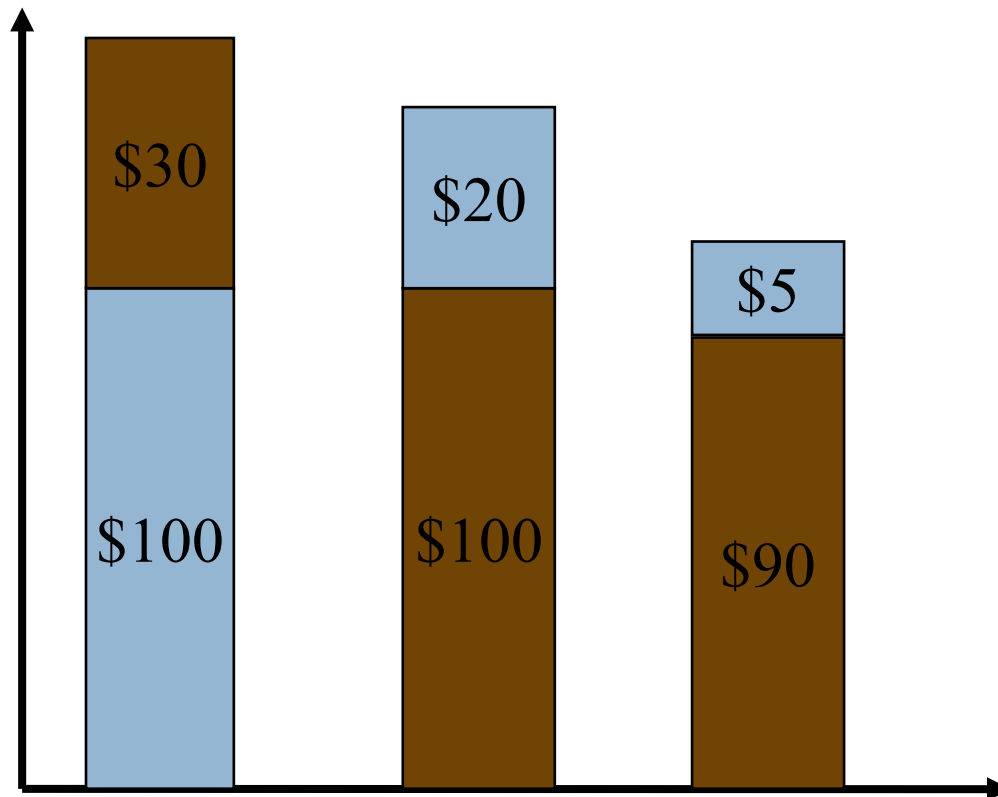
18

Word

Excel

Reporter Analyst Engineer

WTP
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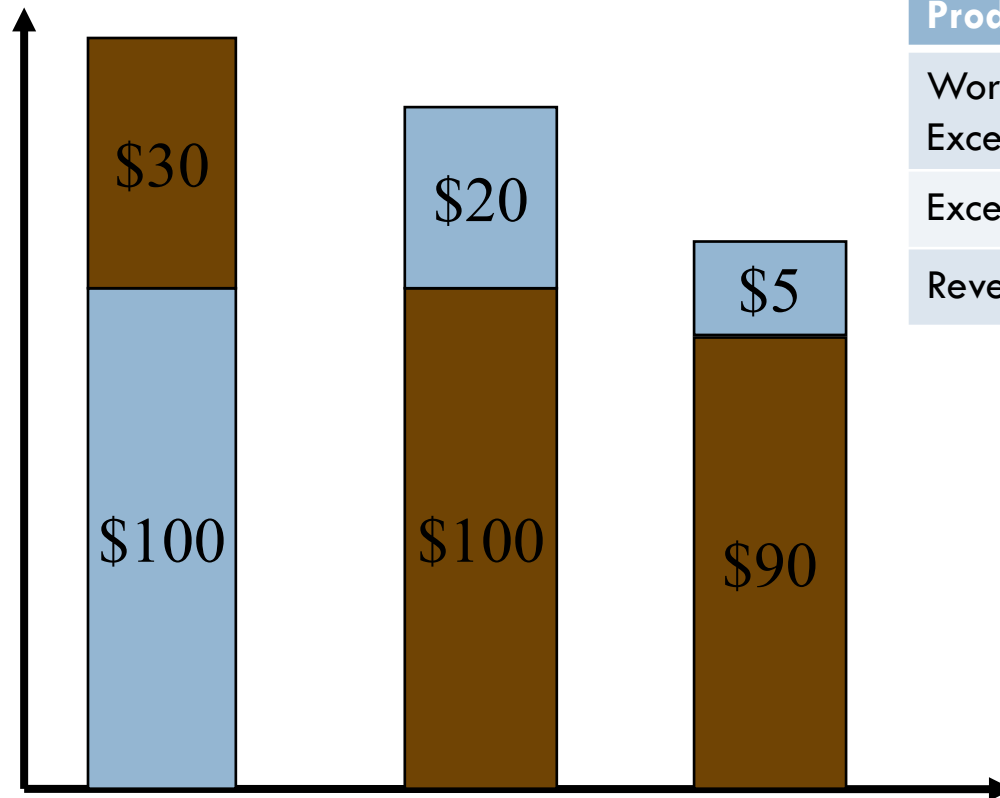
Word

Excel

Analyst will get a \$10 CS if he debundle and buys excel alone.
Solution; must cut the bundle price by \$10

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter
Excel Only	\$ 90	Engineer + Analyst
Revenue	\$300	

Mixed bundling and non-complements

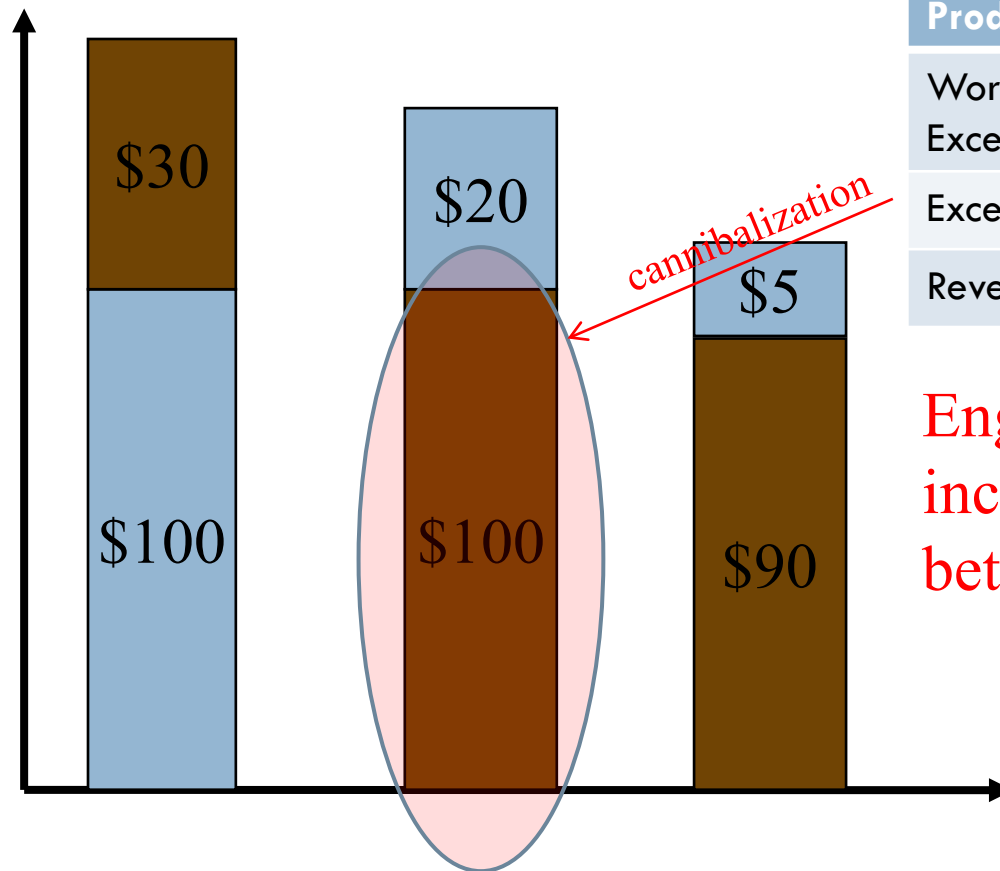
18

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter
Excel Only	\$ 90	Engineer + Analyst
Revenue	\$300	

Engineer profits are incremental. But we can do better.

Mixed bundling and non-complements

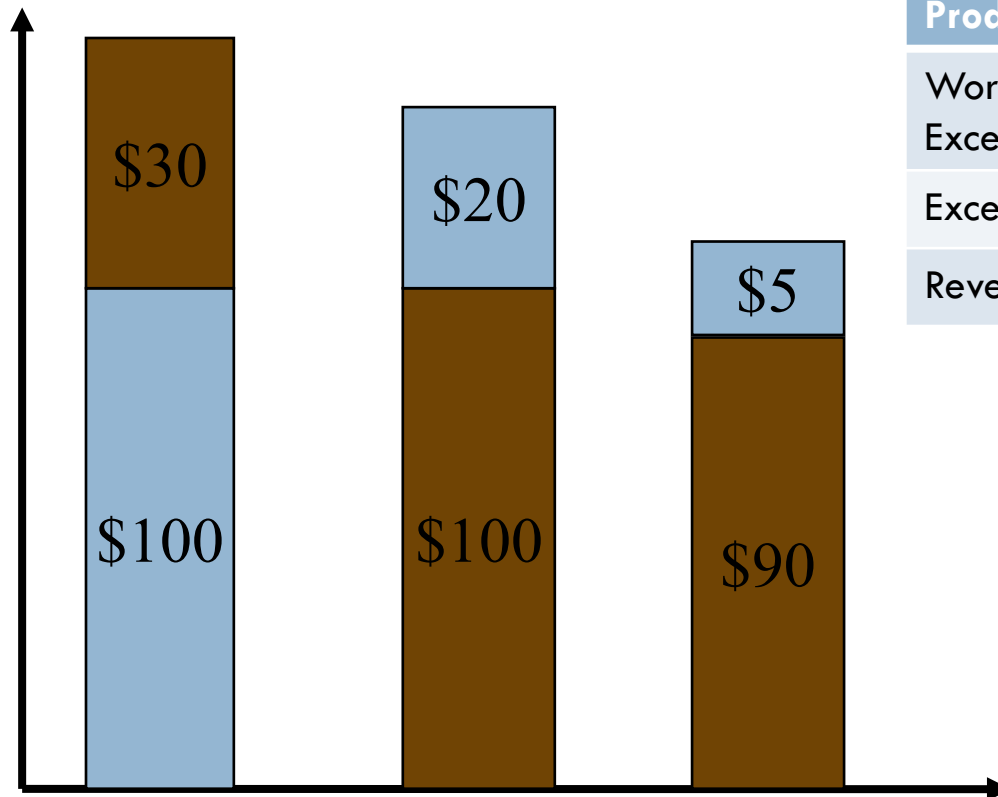
18

Word

Excel

Reporter Analyst Engineer

WTP
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Product	Price	Demand
Word + Excel	\$120	Reporter
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Revenue	\$300	

Mixed bundling and non-complements

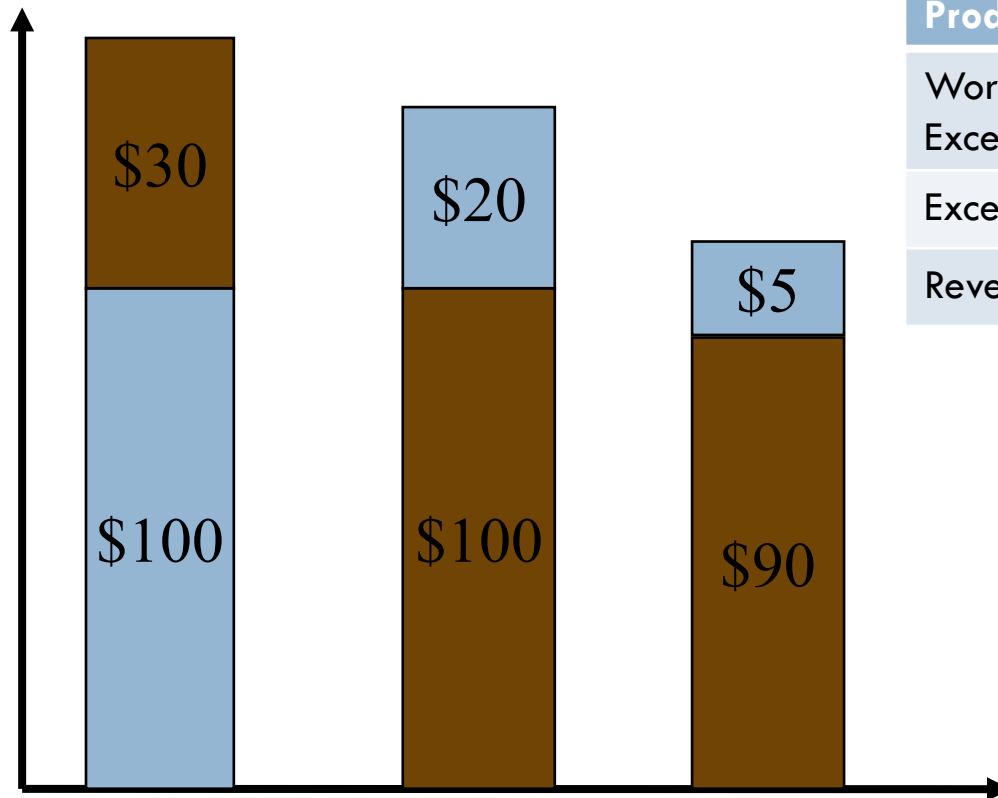
18

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$110	Reporter + Analyst
Excel Only	\$ 90	Engineer
Revenue	\$310	

Mixed bundling and non-complements

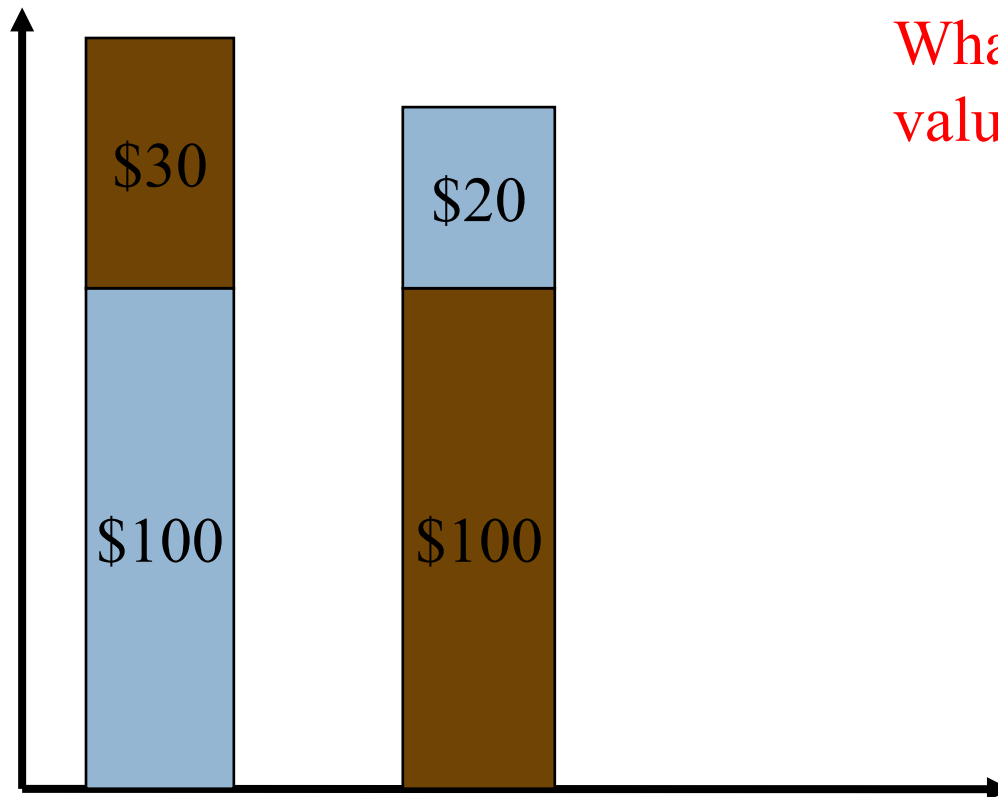
19

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What if the engineer
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Mixed bundling and non-complements

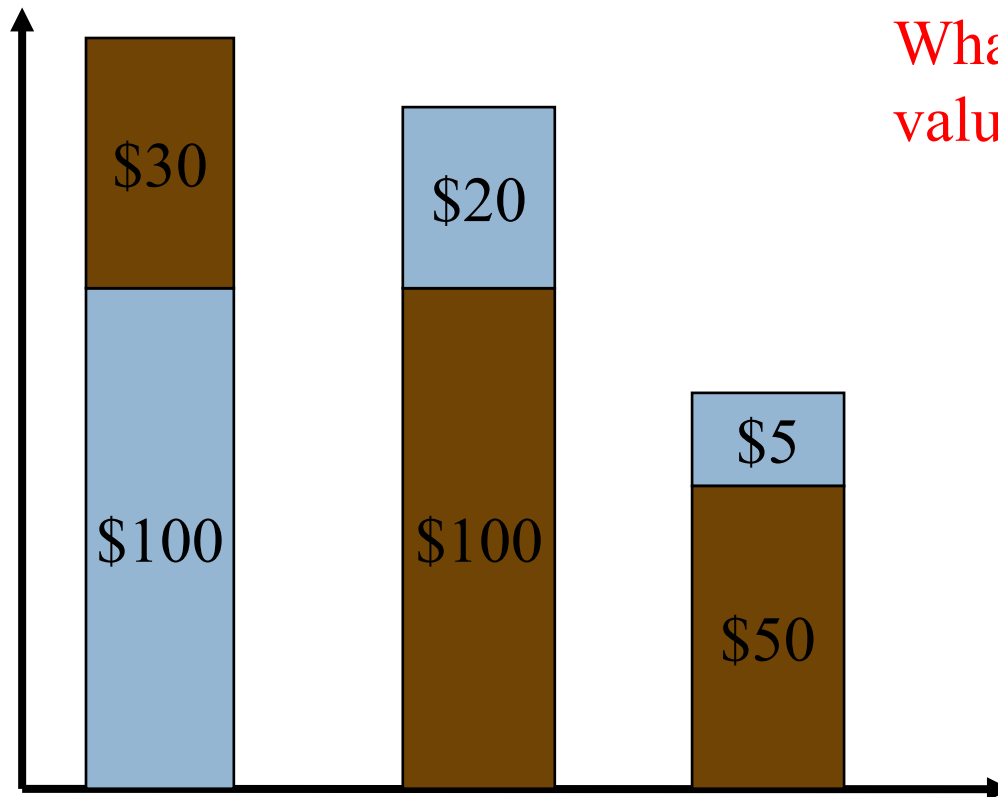
19

Word

Excel

Reporter Analyst Engineer

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Bundle



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Mixed bundling and non-complements

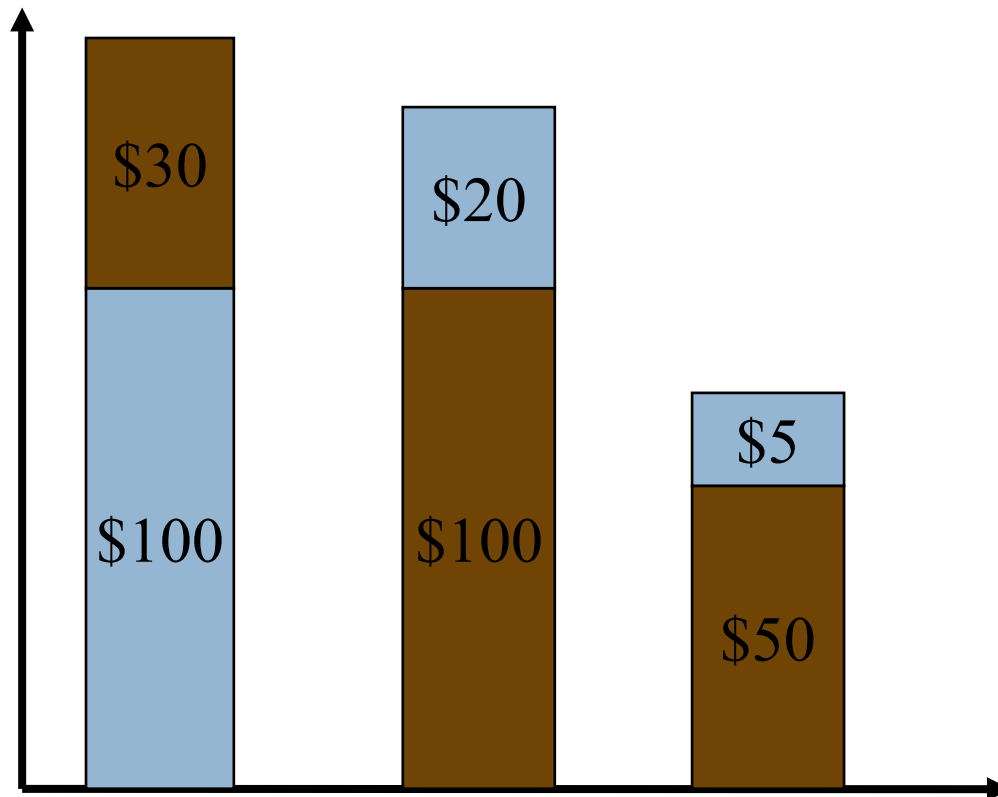
19

Word

Excel

Reporter Analyst Engineer

WTP
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Mixed bundling and non-complements

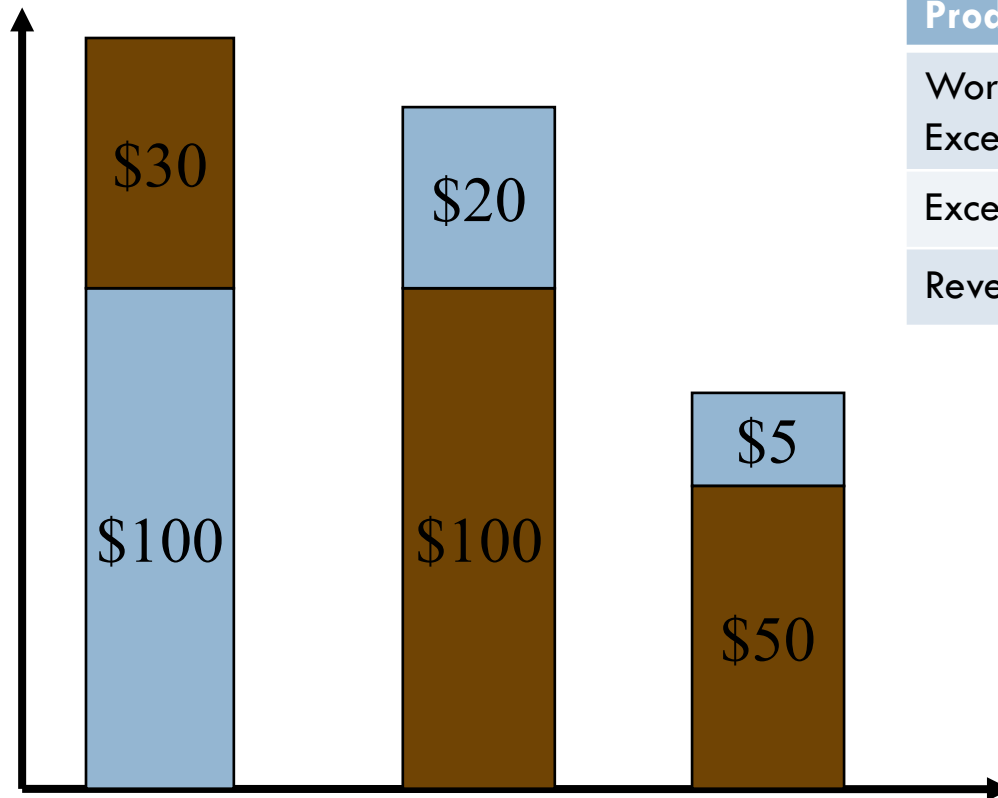
19

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter
Excel Only	\$ 50	Engineer + Analyst
Revenue	\$220	

Mixed bundling and non-complements

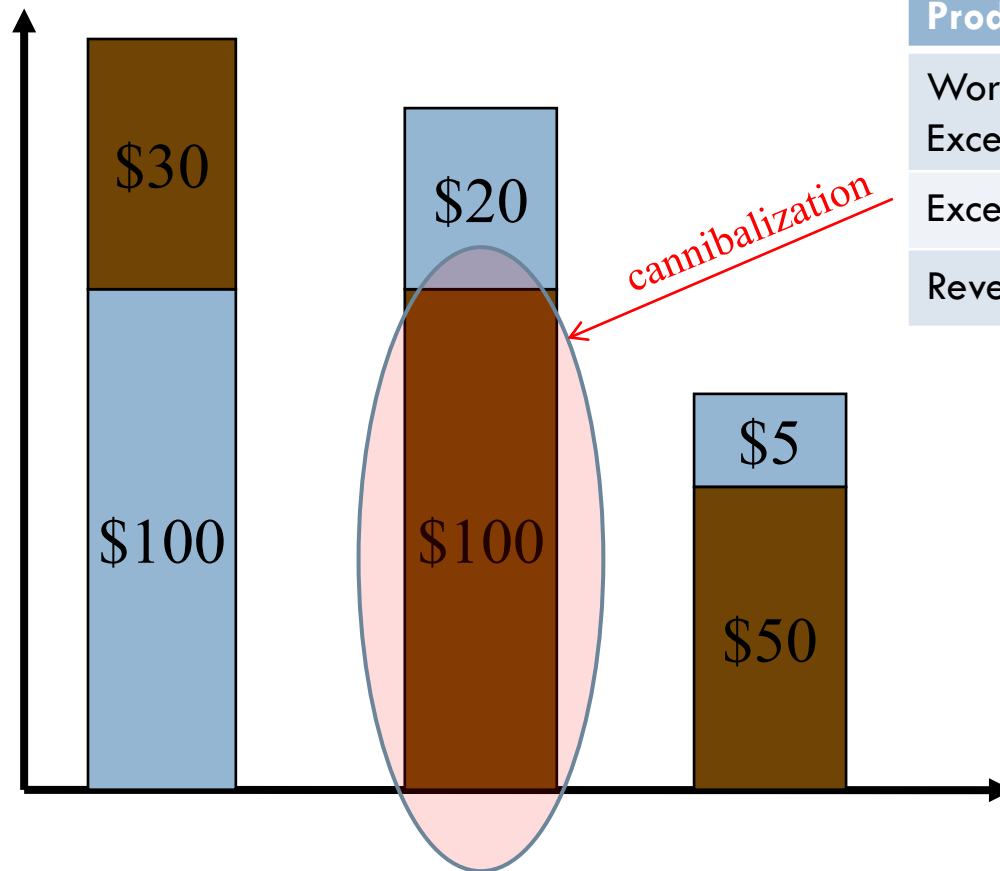
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WTP
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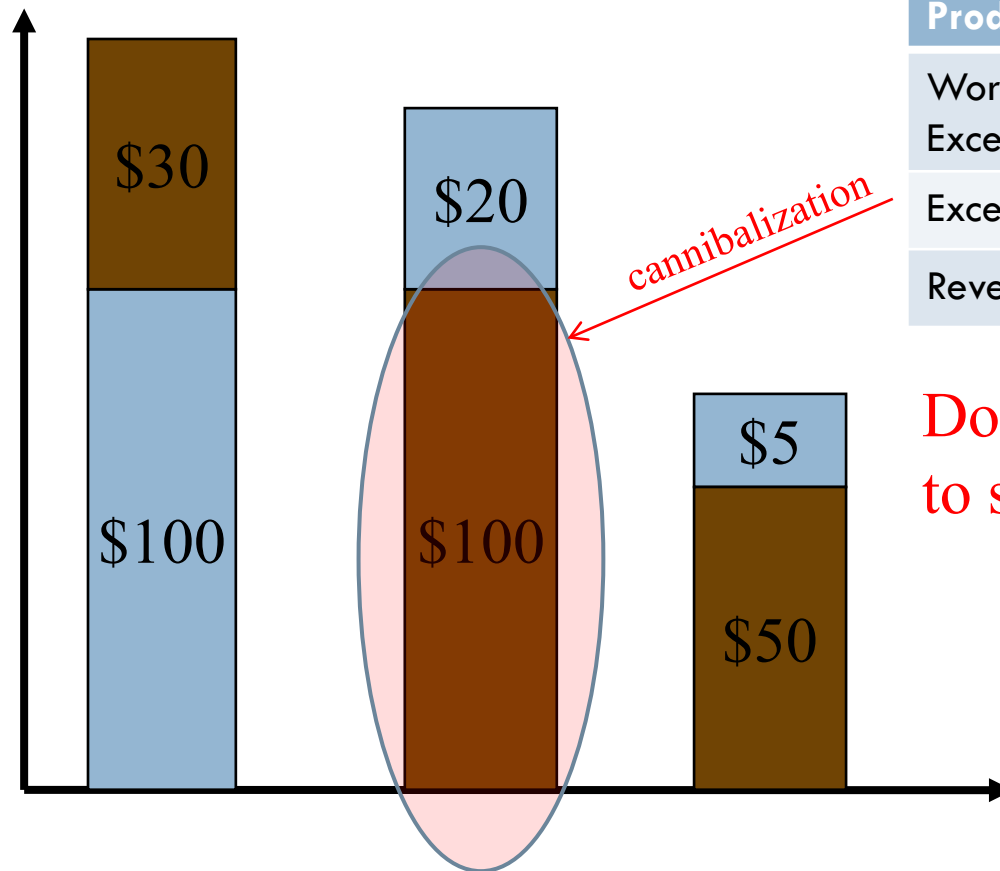
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Word

Excel

Reporter Analyst Engineer

WTP
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Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter
Excel Only	\$ 50	Engineer + Analyst
Revenue	\$220	

Does it really make sense to sell Excel unbundled?

Mixed bundling and non-complements

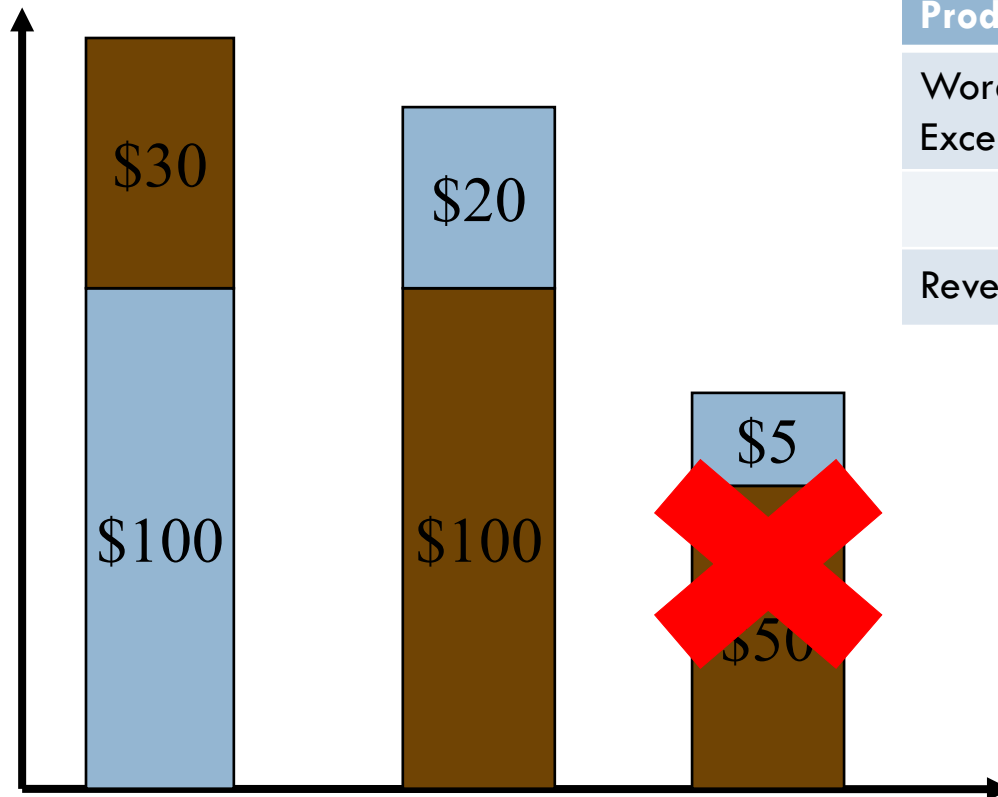
19

Word

Excel

Reporter Analyst Engineer

WTP
For
Bundle



Product	Price	Demand
Word + Excel	\$120	Reporter + Analyst
Revenue	\$240	

Mixed Bundling and non-complements

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- **Mixed Bundling** is just segmentation with a Product Line.
- As before, **coordination** is crucial to avoid **cannibalization**.
 - use individual components (*unbundled*) for new demand in segments that would not buy the bundle.

E.g., sell MS Word *à la carte* for students.

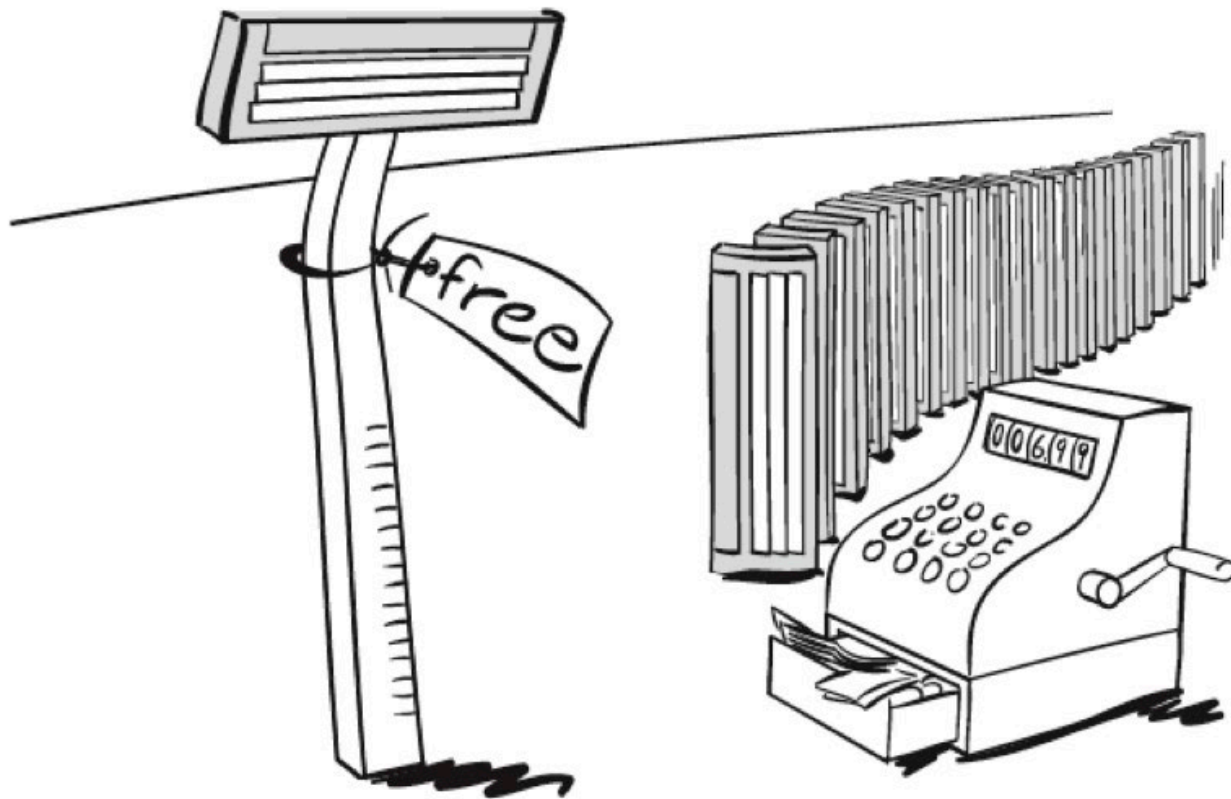
- But, be sure to coordinate the potential cannibalization.

Product line management
requires cannibalisation
management crucially!

- See Appendix for discussion of data analytics and bundling

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Metered Pricing

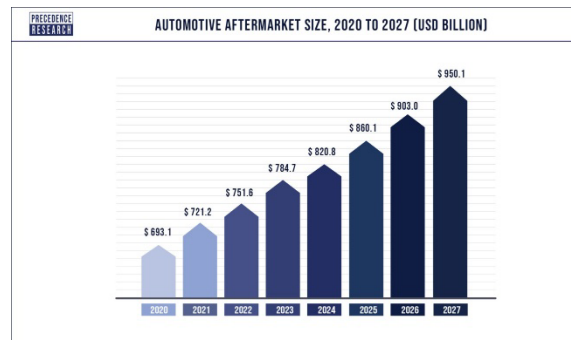


Metered Pricing: The *Razor & Blades* Model

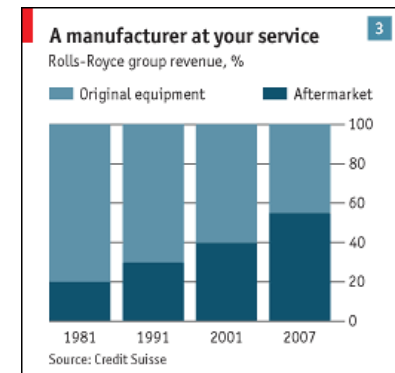
22

- Also known as “*bait-and-hook*” pricing
- Examples:
 - ▣ Gillette Mach 3 Razor \$7.69, 12 blades (3-month supply) \$21.99
 - ▣ Inkjet printers under \$200, HP replacement cartridge \$52
 - ▣ Jet engines cheap (Rolls Royce, GE...), high mark-ups on aftermarket maintenance and replacement parts
 - ▣ Automobile aftermarket also growing fast!!!

Rolls;
engines are elastic (buyers
might look for cheaper
price).
but maintenance services
are very inelastic because
customers locked in.



Globenewswire, Jan. 3, 2022



Economist, Jan. 8, 2009

Metered Pricing Theory

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- Two complementary goods
 - ▣ One with fixed-quantity demand
 - ▣ One with variable-quantity demand
- *Loss-leader* pricing on the fixed-quantity product, then *Price-discriminate* on variable-quantity product
 - ▣ Particularly effective if variable-quantity demand good is a commodity (i.e., hard to profit on it as a stand-alone business)

Metered Pricing Theory

Example: Printers & Ink

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- Suppose you are selling printers and ink cartridges
 - ▣ No competition on printers
 - ▣ high competition on ink (i.e., no **differentiation value** for ink)
 - i.e., stand-alone ink profits close to zero
 - ▣ Marginal Costs: \$80 (printer), \$1.5 (ink cartridge)

If sell just ink (w/o printer),
zero profit (for this
scenario)

Consumer Type	Valuation Per Unit					
	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
2	\$140	\$14	\$12	\$7	0	

Profits and *Razor & blades* theory

Scenario 1 : Printers & ink operated *independently*

25

- Profits only from printers b/c competitive ink market.

Consumer Type	Valuation Per Unit					
	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
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Profits and *Razor & blades* theory

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- If you charge \$210 for printer

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Profits and *Razor & blades* theory

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- Profits only from printers b/c competitive ink market.
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Profits and *Razor & blades* theory

Scenario 1 : Printers & ink operated *independently*

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- Profits only from printers b/c competitive ink market.
- If you charge \$210 for printer
 - ▣ Sell only to high WTP customers
 - ▣ Total profits: \$130 ← $\text{Profit} = (\$210 - \$80) \times 1 = \$130$

Consumer Type	Valuation Per Unit					
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1	\$210	\$16	\$14	\$10	\$5	0
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- If you charge \$140 for printer

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	Fixed-Quantity Good	Variable Quantity Good				
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- If you charge \$210 for printer
 - ▣ Sell only to high WTP customers
 - ▣ Total profits: \$130 ← $\text{Profit} = (\$210 - \$80) \times 1 = \$130$
- If you charge \$140 for printer
 - ▣ Sell to both high and low WTP customers

Consumer Type	Valuation Per Unit					
	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
2	\$140	\$14	\$12	\$7	0	

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- If you charge \$210 for printer
 - ▣ Sell only to high WTP customers
 - ▣ Total profits: \$130 ← $\text{Profit} = (\$210 - \$80) \times 1 = \$130$
- If you charge \$140 for printer
 - ▣ Sell to both high and low WTP customers
 - ▣ Total profits: \$120

Consumer Type	Valuation Per Unit					
	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
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Profits and *Razor & blades* theory

Scenario 1 : Printers & ink operated *independently*

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- Profits only from printers b/c competitive ink market.

- If you charge \$210 for printer
 - ▣ Sell only to high WTP customers
 - ▣ Total profits: \$130

$$\text{Profit} = (\$210 - \$80) \times 1 = \$130$$

- If you charge \$140 for printer
 - ▣ Sell to both high and low WTP customers
 - ▣ Total profits: \$120

Consumer Type	Valuation Per Unit					
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		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
2	\$140	\$14	\$12	\$7	0	

Profits and *Razor & blades* theory

Scenario 2: Printers & ink *tied together*

26

- Tie ink cartridges to printer to **lock-in** ink sales via printer sales

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	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
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Profits and *Razor & blades* theory

Scenario 2: Printers & ink *tied together*

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- If you charge \$210 for printer

Consumer Type	Valuation Per Unit					
	Fixed-Quantity Good	Variable Quantity Good				
		1	2	3	4	5
1	\$210	\$16	\$14	\$10	\$5	0
2	\$140	\$14	\$12	\$7	0	

Profits and *Razor & blades* theory

Scenario 2: Printers & ink *tied together*

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- If you charge \$210 for printer

- Only sell to high WTP customers

- Optimal price for cartridges is \$10

- Total profits: \$130 + \$25.5 = \$155.5

Ink Profit

$$= (\$10 - \$1.5) \times 3 = \$25.5$$

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Profits and *Razor & blades* theory

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 - *loss-leader pricing* to leverage monopoly on base good
- **Price segmentation:** Consumer segment with high WTP for printer also buys more ink and spends more!

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Can still blend in NL pricing methods such as Volume Discount or ink bundling to further maximise profits.

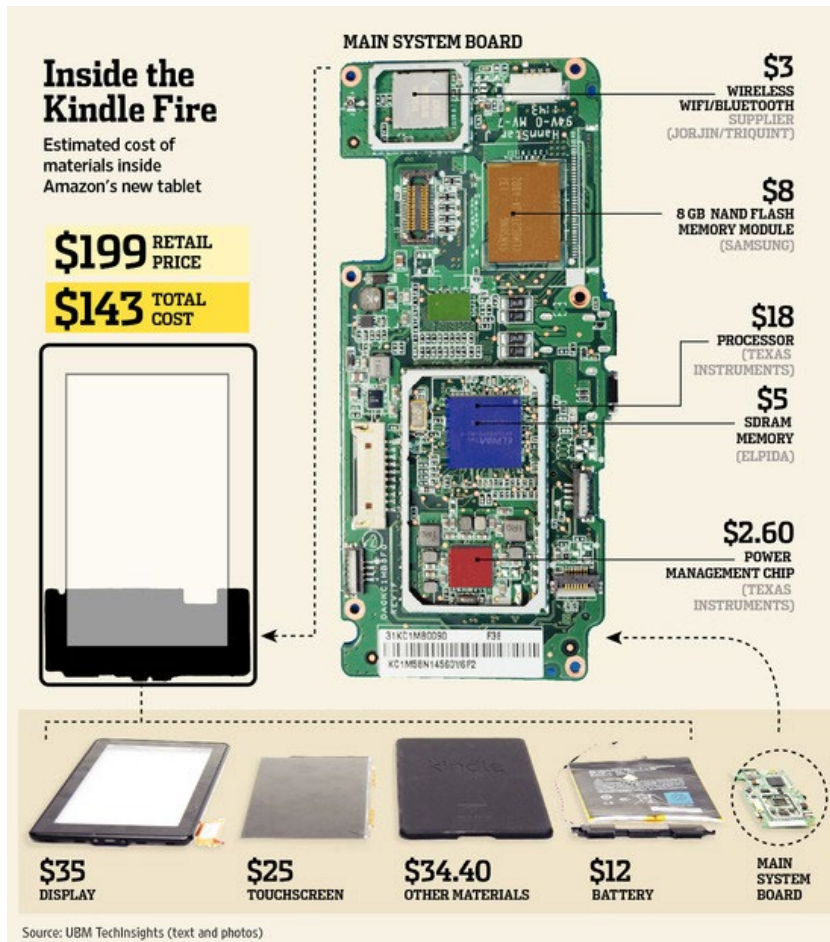
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Exercise: See if you can re-do scenario 2 with an even more profitable volume discount on the variable-quantity good.

Metering over time

Example: Amazon's Kindle Fire in fall 2011

28



- Price is \$199
 - versus \$499 for iPad
- Total unit cost \$143 (\$203 if include manuf. services)
- How can this strategy be profitable?

"...Amazon mainly hopes to make money from sales of digital media and physical goods that consumers may order from Amazon on the device..."

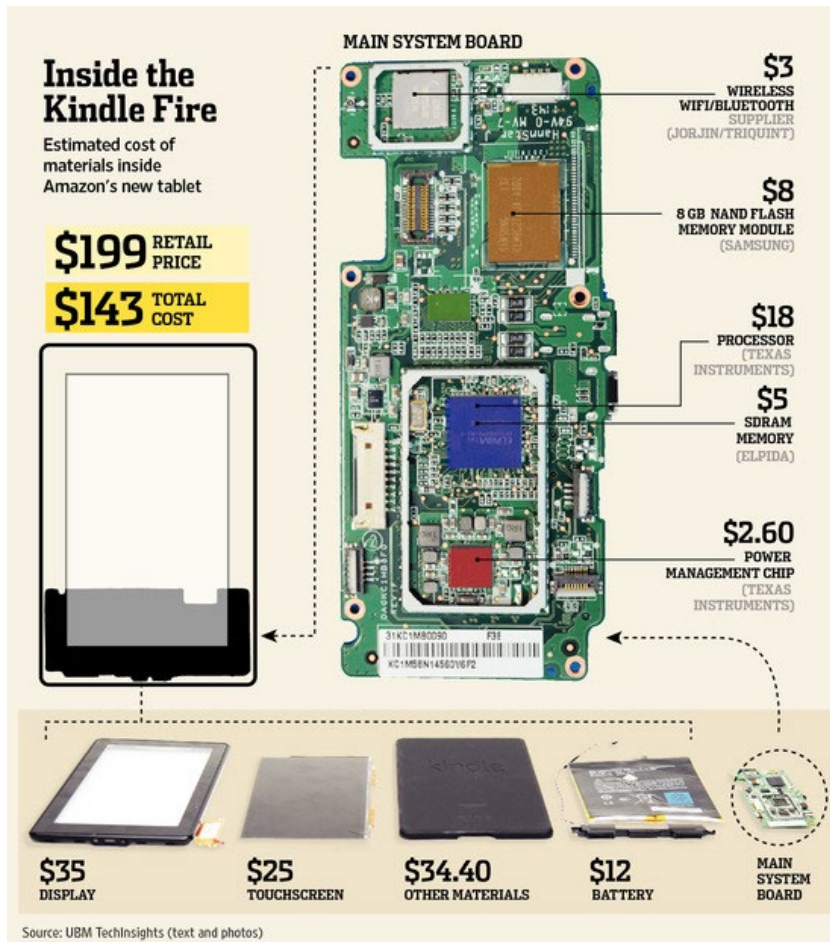
"...the device will serve as a platform for the sale of content..."

WSJ 11-18-2011

Metering over time

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- How can this strategy be profitable?

“...Amazon from sale goods to Amazon

“We sell the hardware at our cost, so it is break-even on the hardware.”

“...the sale of

WSJ 11-18-2011

Jeff Bezos, 9-12-2012

B2B Metered Pricing

Cloud Computing

29

- Heavy investment in cloud services (Q1 2015):

-  Amazon: \$871 mill

-  Microsoft: \$1.4 bill

-  Google: \$2.93 bill

“Silver Linings in Big Cloud Spending,” WSJ, 4-27-2015

- Cloud storage seems commoditized

Its not about storage. storage is just the razor. the blades is the main profit center

- But ...

- Charge each time data accessed/processed

- Data management services

- “Rent” software in cloud (SaaS)

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generated \$56.5 Billion in 2023
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Cloud Computing

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□ Heavy investment



Sustainable Growth?

- ...another notable slowdown in growth for their respective cloud-computing services following a weak December quarter as large companies continue to scrutinize their spending on technology...
- all three have soundly outperformed the S&P 500...
- another key facet is hype over **generative artificial intelligence** technology that all three are incorporating into their services...

"Big Tech Depends on AI Shining Through Cloud Haze" WSJ, April 23, 2023

4-27-2015

□ Cloud

□ But ...

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B2B Metered Pricing Cloud Computing

30

Similar story playing out in
China

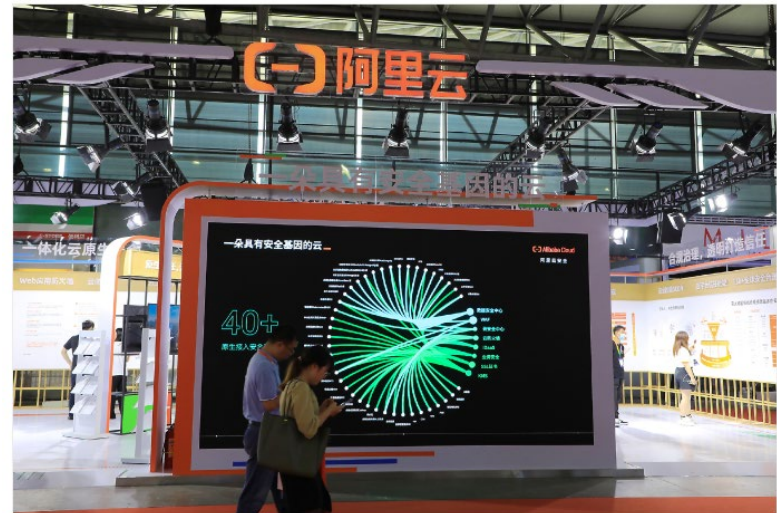
“Expenditure on cloud
infrastructure services on the
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to \$9.2 billion in Q3 2023...”

“...Alibaba Cloud's price cuts
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CHINADAILY 中国日报网
CHINADAILY.COM.CN

Alibaba price cuts heat up cloud space

By FAN FEIFEI | China Daily | Updated: 2024-03-02 09:30



B2B Metered Pricing Cloud Computing

30

Price War?
Or
Razors & Blades?!

Similar story playing out in
China

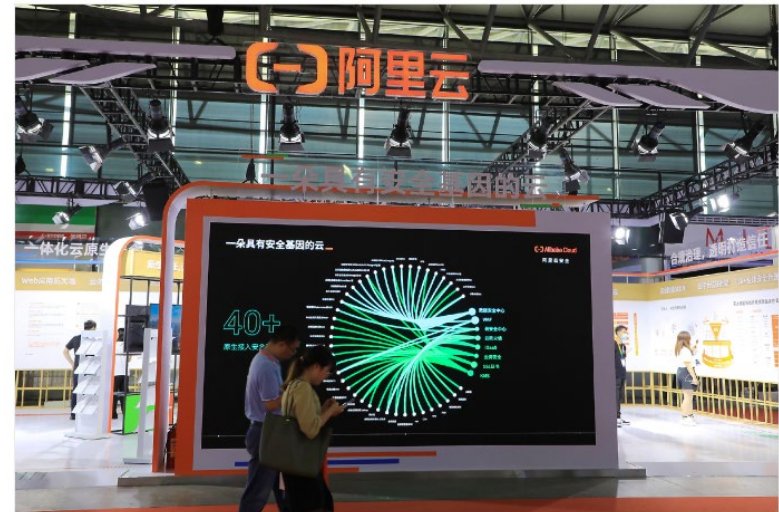
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CHINADAILY 中国日报网
CHINADAILY.COM.CN

Alibaba price cuts heat up cloud space

By FAN FEIFEI | China Daily | Updated: 2024-03-02 09:30



Metering as **Marketing For Good**

Wi-fi in India

31

Two Years Ago, India Lacked Fast, Cheap Internet—One Billionaire Changed All That

Mukesh Ambani's national 4G network for Reliance Jio, which charges pennies for data, is bringing hundreds of millions online for the first time

WSJ, 5-5-2018

- ▣ Free calls
- ▣ Data for pennies
- ▣ Opens market to 1.3 billion people
- ▣ **Company's plan:**



“use the platform to sell content, financial services and advertising.”

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advertisi



471 million active users!
₹187 ARPU (about \$2.2 USD)

→ 11% EBITDA growth

Mint, 01-12-2024

“Reliance Jio will move India
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Mukesh Ambani

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Glenn Chapman, *Phys.org*, 7-20-2018

Threats to Metered Pricing

32

- Release of Compatible Generics into secondary ink market
 - ▣ Staples has generic ink cartridges for 25% less
 - ▣ Inkjet refill kits \$15
 - ▣ Resale of refurbished (i.e., refilled) ink cartridges



Home / Lexmark / Remanufactured Lexmark 20 (15M0120) Color Ink Cartridge - 2 Pack



Remanufactured Lexmark 20 (15M0120) Color Ink Cartridge - 2 Pack

★★★★★ 1 Review

\$27.89

~~\$81.90~~ (Save 66%)

− 0 +

ADD TO CART

- Color: **Color**
- Capacity: **Standard Yield**
- Page Yield: **275 pages each**

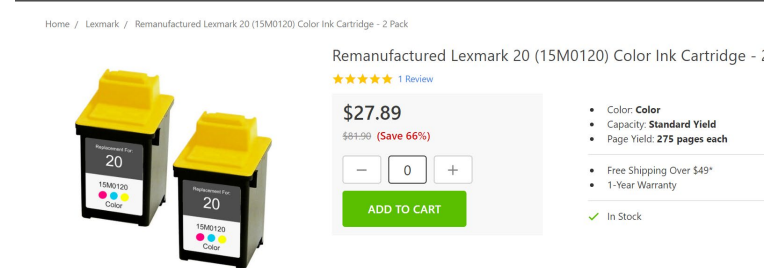
- Free Shipping Over \$49*
- 1-Year Warranty

✓ In Stock

Threats to Metered Pricing

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- Lexmark blocked re-fill kits with sensor
- Lexmark recently fought to prevent resale of used cartridges

Threats to Metered Pricing

33

- Antitrust Concerns
- Legality of Tying (protection of intellectual property versus foreclosure of rivals)
- Some famous cases:
 - ▣ Microsoft
 - ▣ Lexmark
 - ▣ iTunes and iPod in Europe (interoperability)

Take Away

34

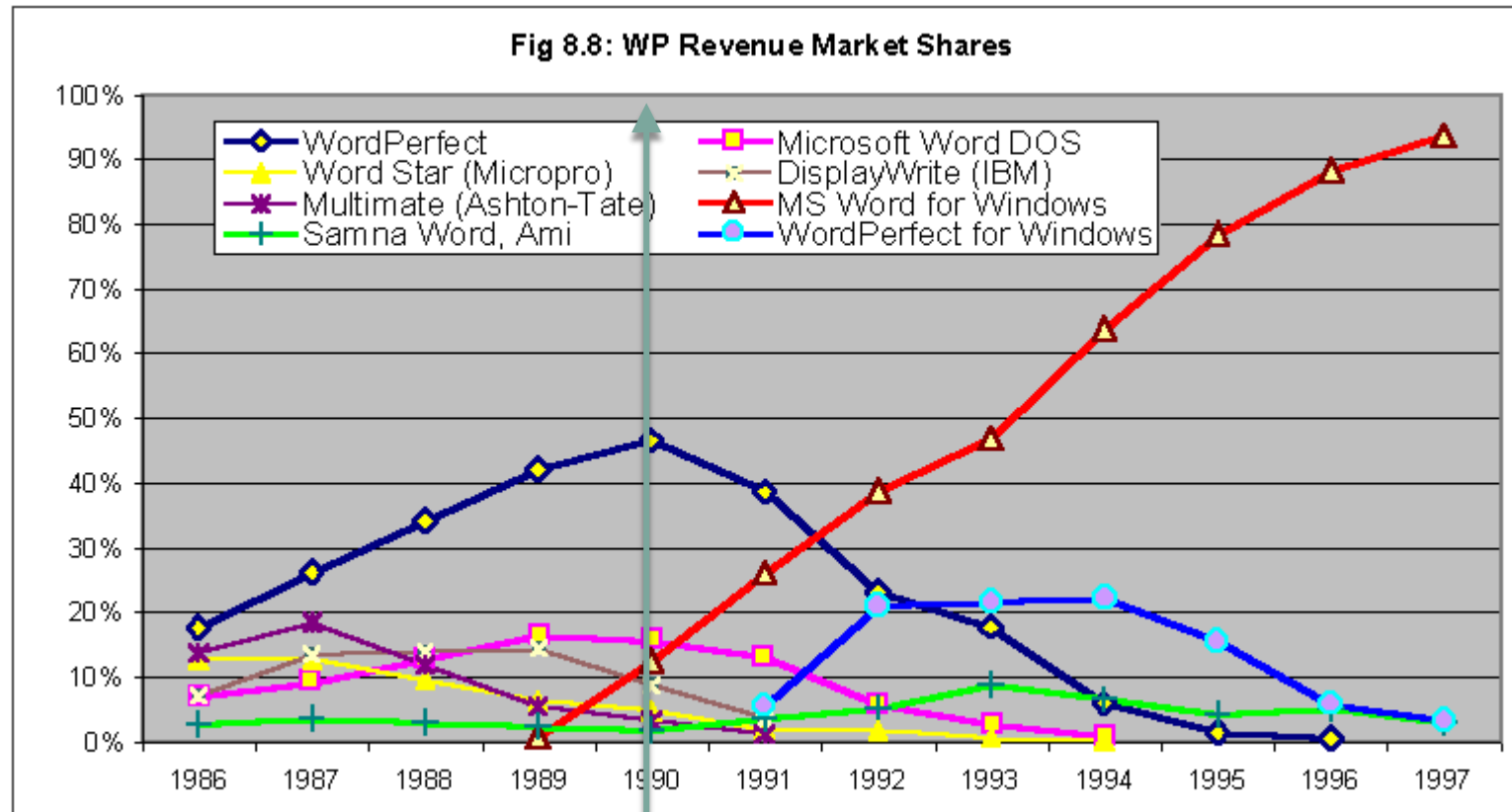
- **Bundling & Metering** as price discrimination (capture value)
- These are Product Line Pricing Strategies
 - **Mixed Bundling** and substitutes
 - **Metering** and complements (leads to *loss-leading* on the “bait” product)
- Coordination of **P**rice and **P**roduct again...
- Legality of **Metering** can be tricky
 - Illegal as means to foreclose competition
- The Economics of **inattention** raises new ethical/legal considerations

35

Microsoft Office Software

MS Word take-off in the U.S.

36



MS Office launch in 1990

37

More examples

Investing in Retention

38

AT&T, Verizon Push Free iPhones for Long-Term Customers

WSJ, June 5, 2021

- Subsidized phones for trade-ins from **new** & **existing** subscribers
- New iPhone subsidies on \$1,000 price: from free to \$700 rebate

Investing in Retention

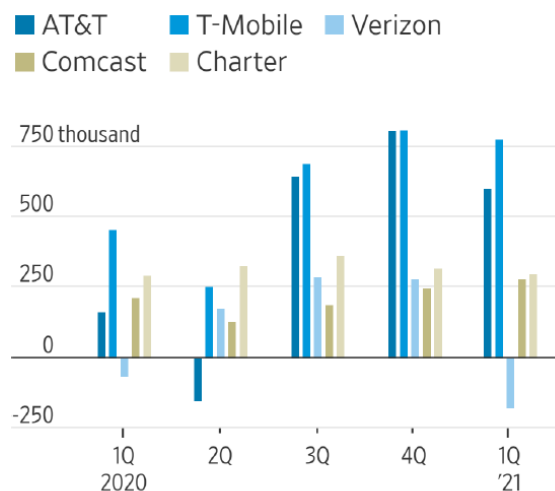
38

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Net additions, postpaid phones



Note: Sprint merged with T-Mobile April 1, 2020

Source: UBS, the companies

Investing in Retention

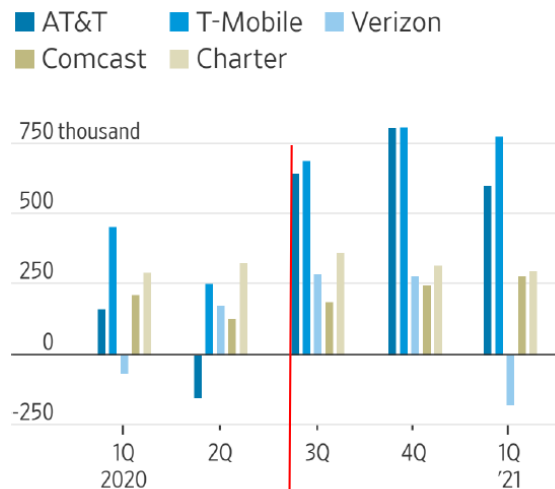
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AT&T: free iPhone 12

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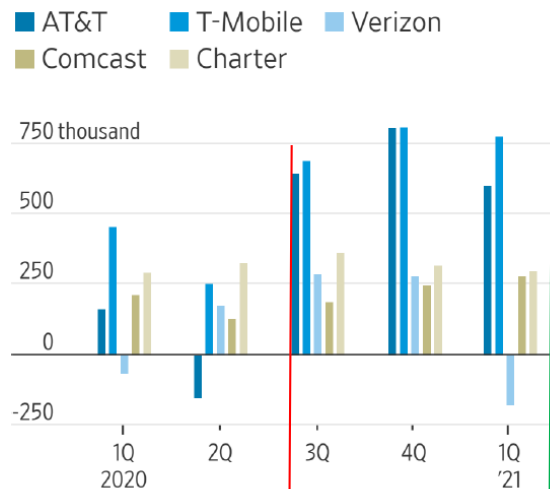
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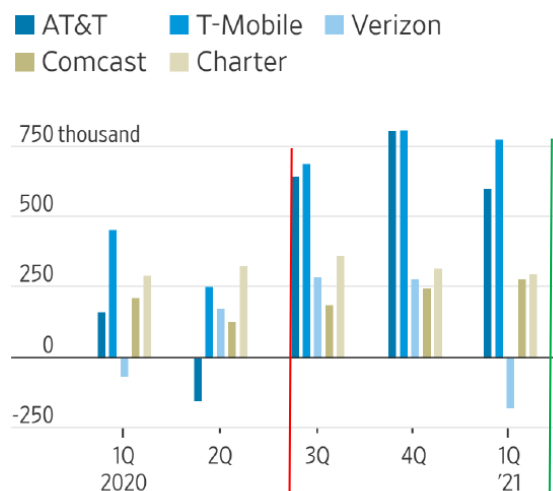
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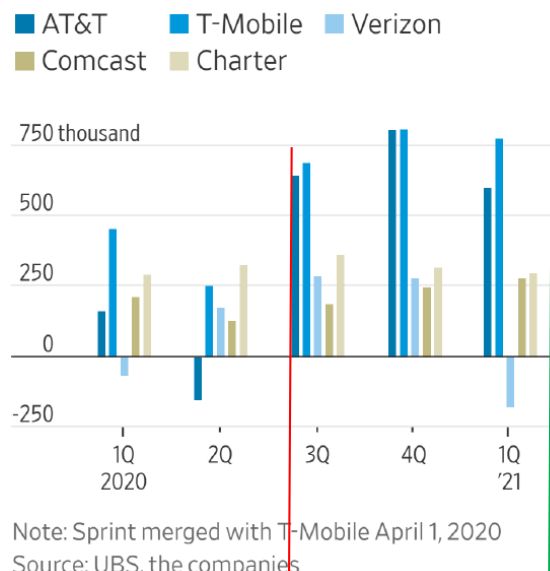
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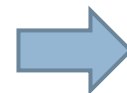
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AT&T added 1.4 million post-paid connections (Oct '20 –March'21)

AT&T: free iPhone 12

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Extra

Data analytics and Bundling

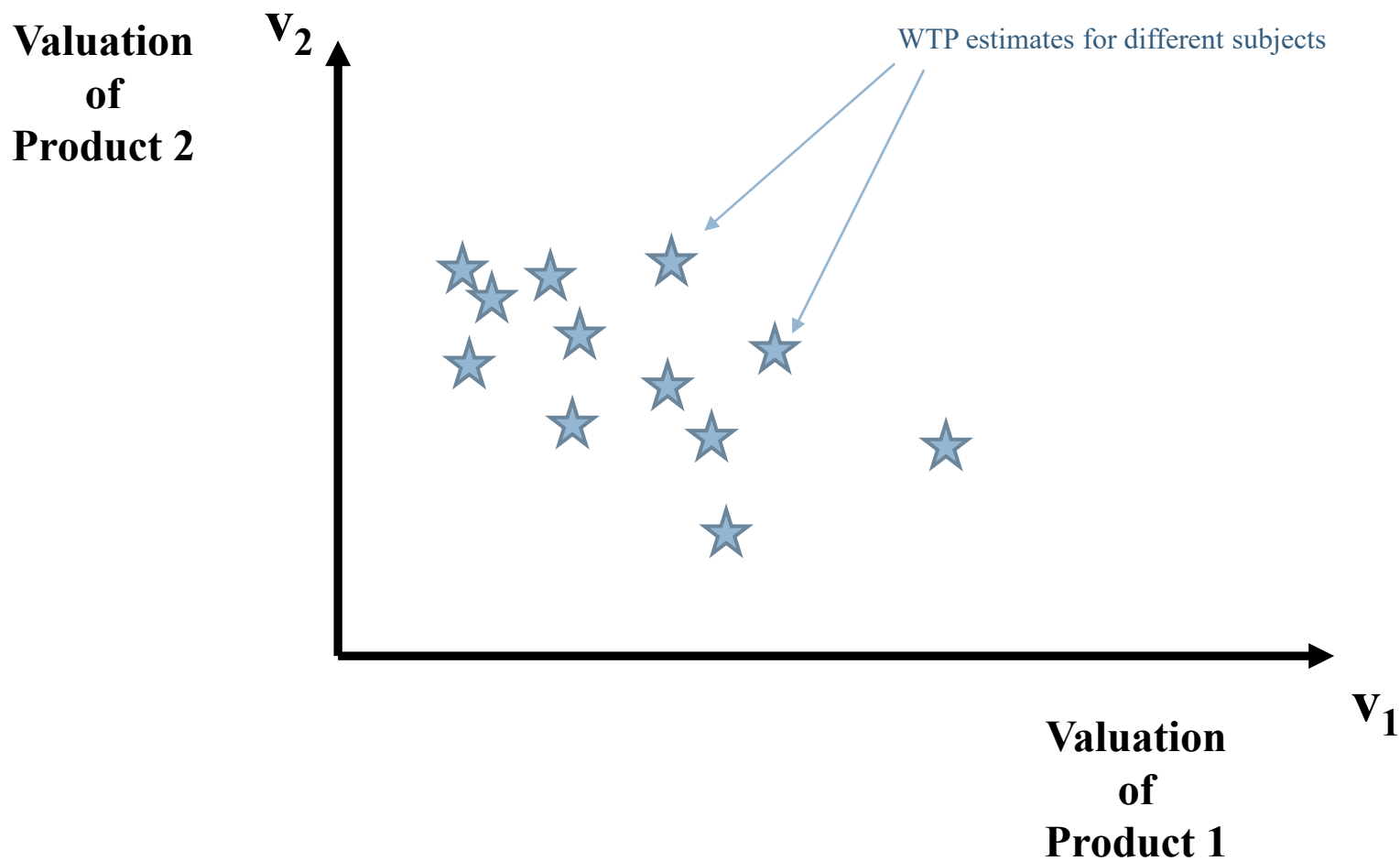
Data analytics and bundling

40

- Let's consider the data analytics associated with bundling
- What would we need to observe in our marketing research to suggest a motive for bundling non-complements?
- Suppose you have conducted a conjoint analysis and estimated consumer valuations (WTP) for your components...

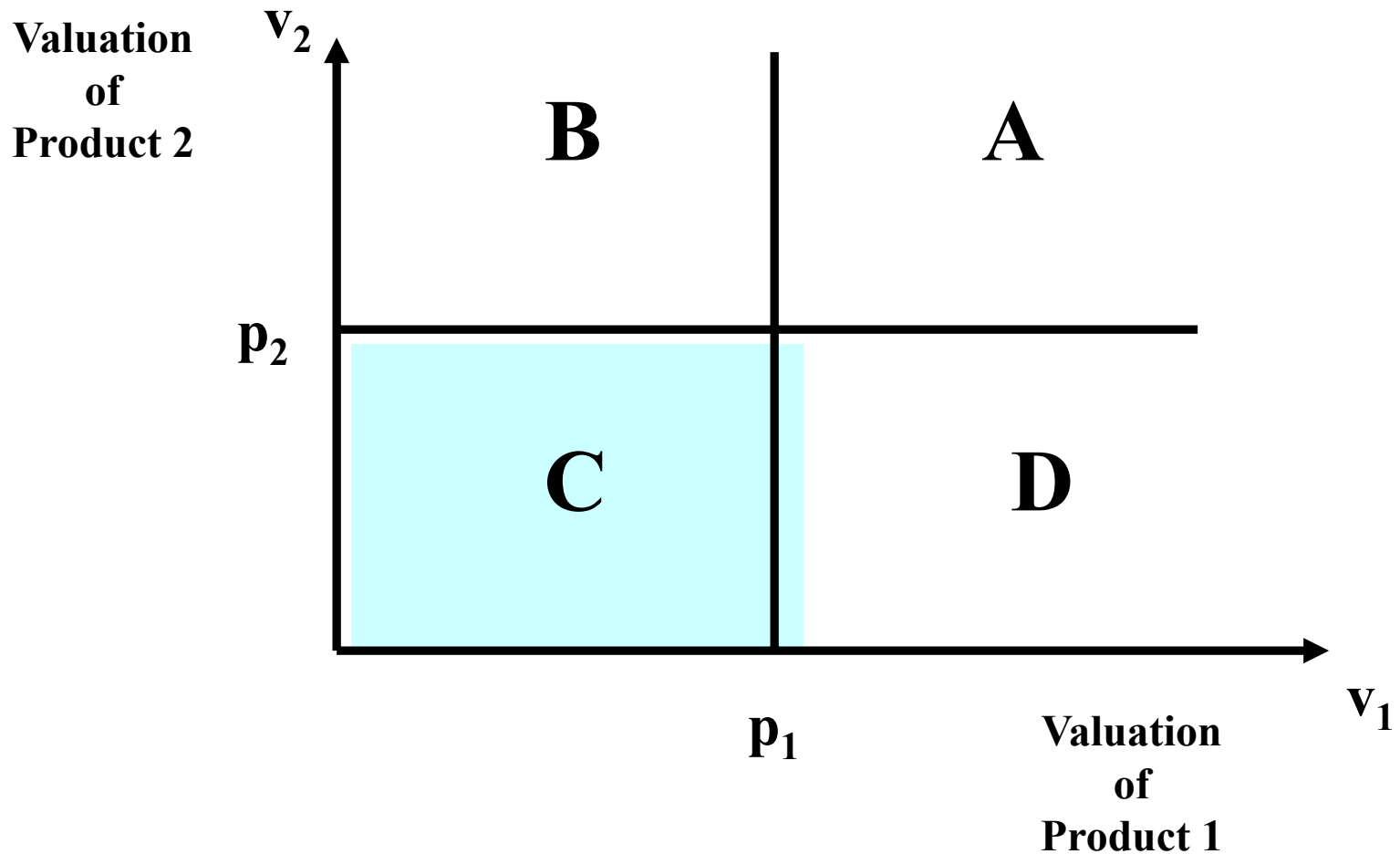
Data

41



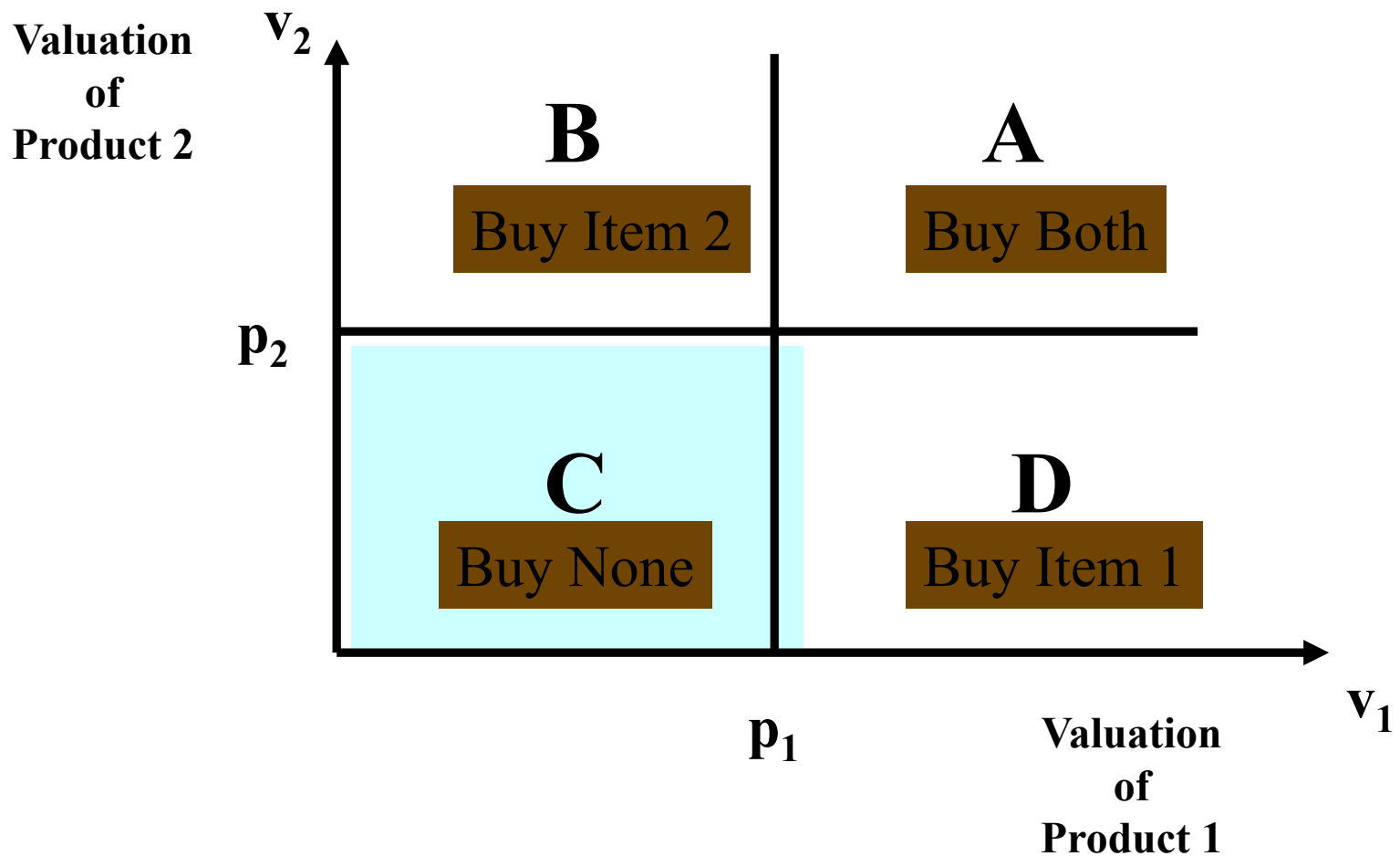
Pure Components Strategy

42



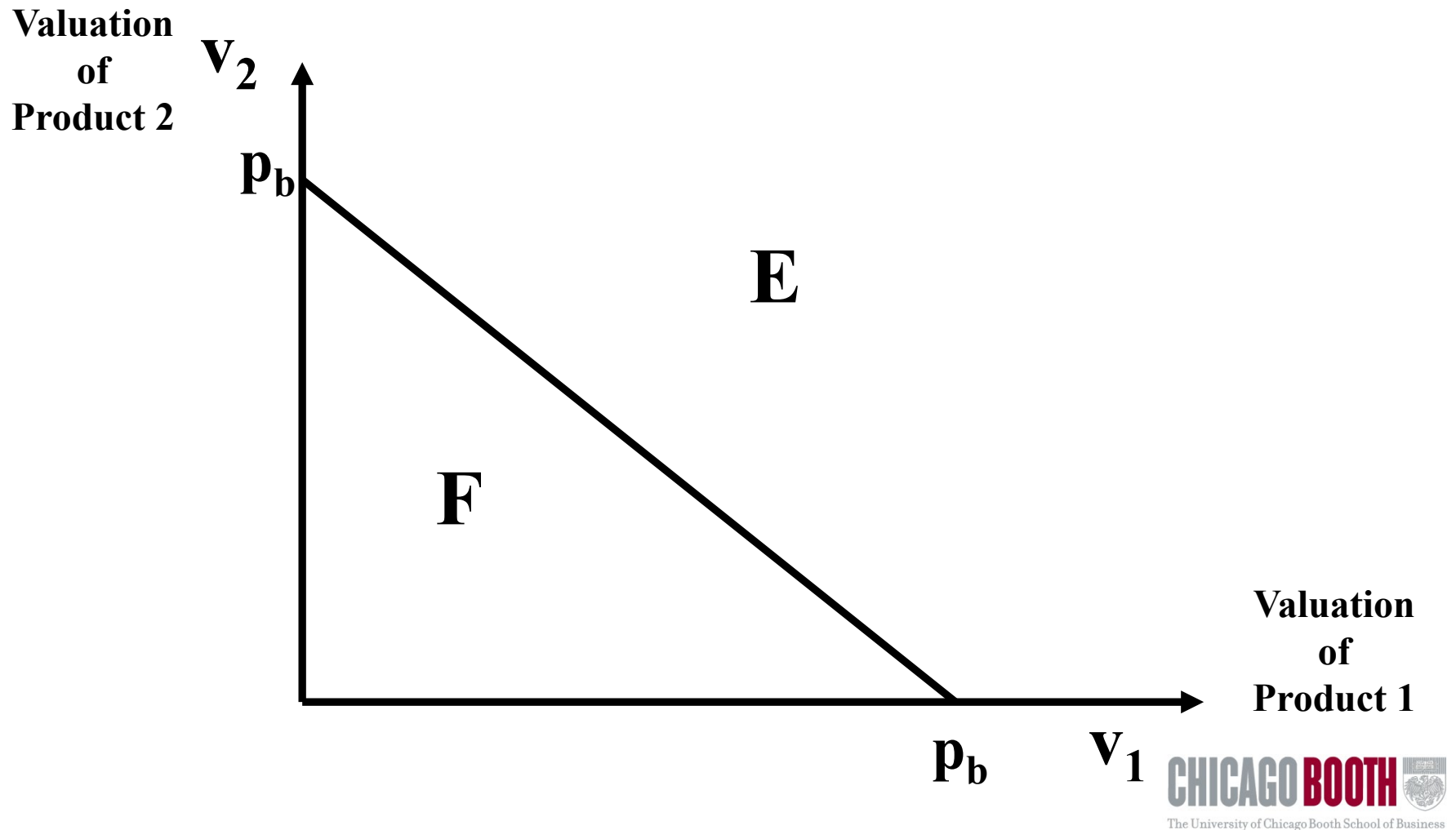
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42



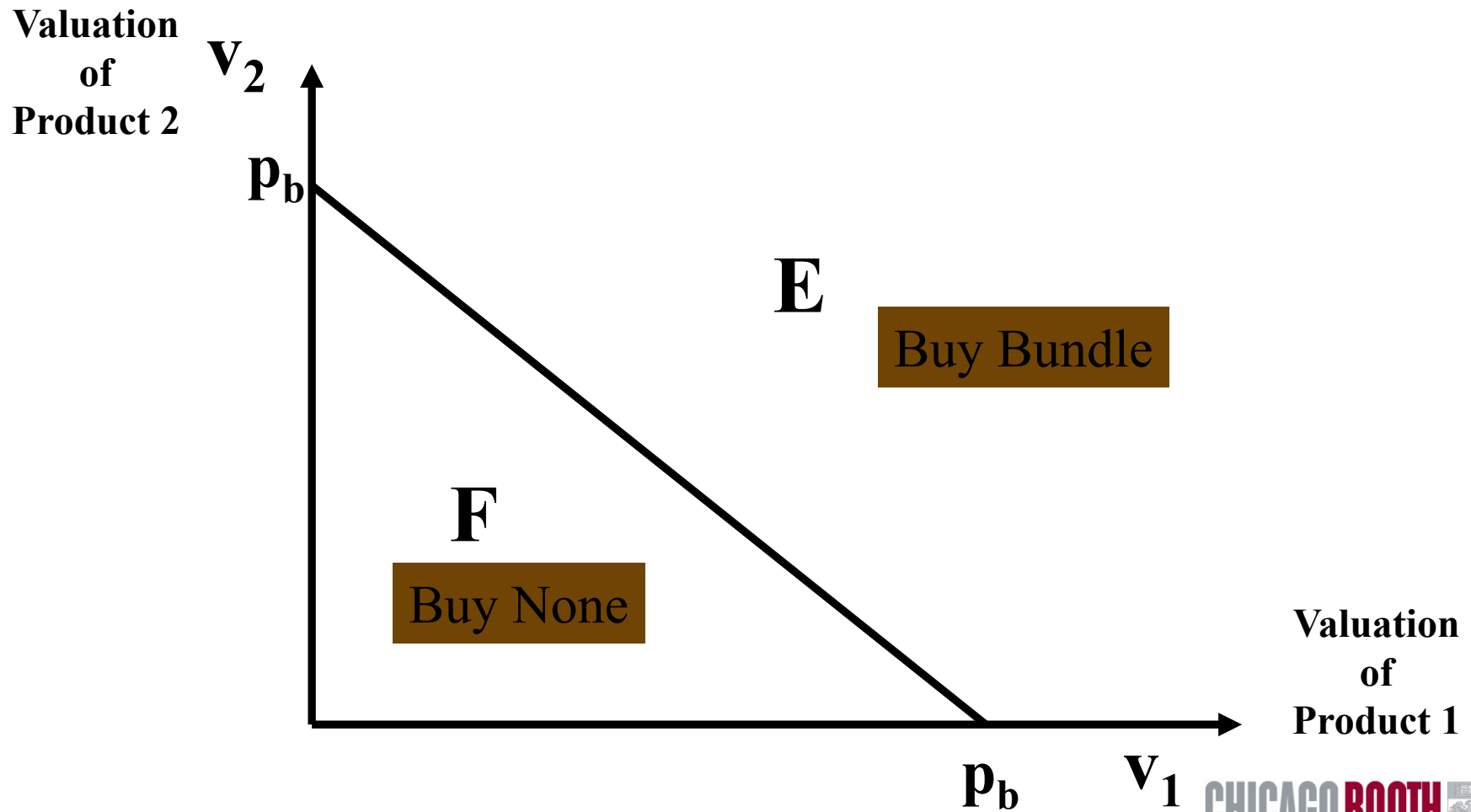
Pure Bundling Strategy

43



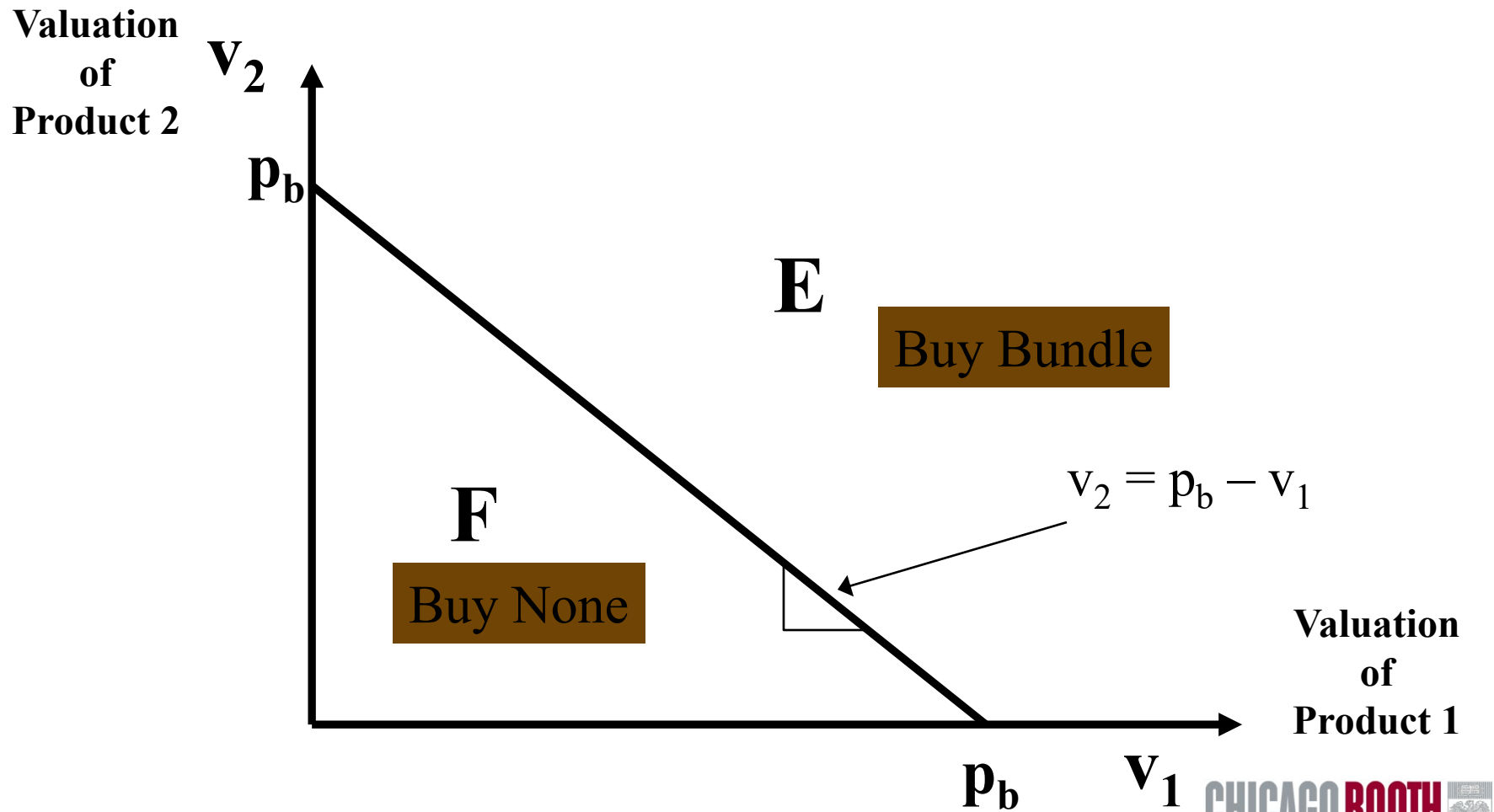
Pure Bundling Strategy

43



Pure Bundling Strategy

43



Pure Bundling Strategy

43

If bundle was 4 units of product 2 and 1 unit of product 1, then:

$$4v_2 + v_1 = p_b \rightarrow v_2 = p_b/4 - v_1/4$$

Valuation
of
Product 2



E

Buy Bundle

F

Buy None

$$v_2 = p_b - v_1$$

Valuation
of
Product 1

p_b

v_1

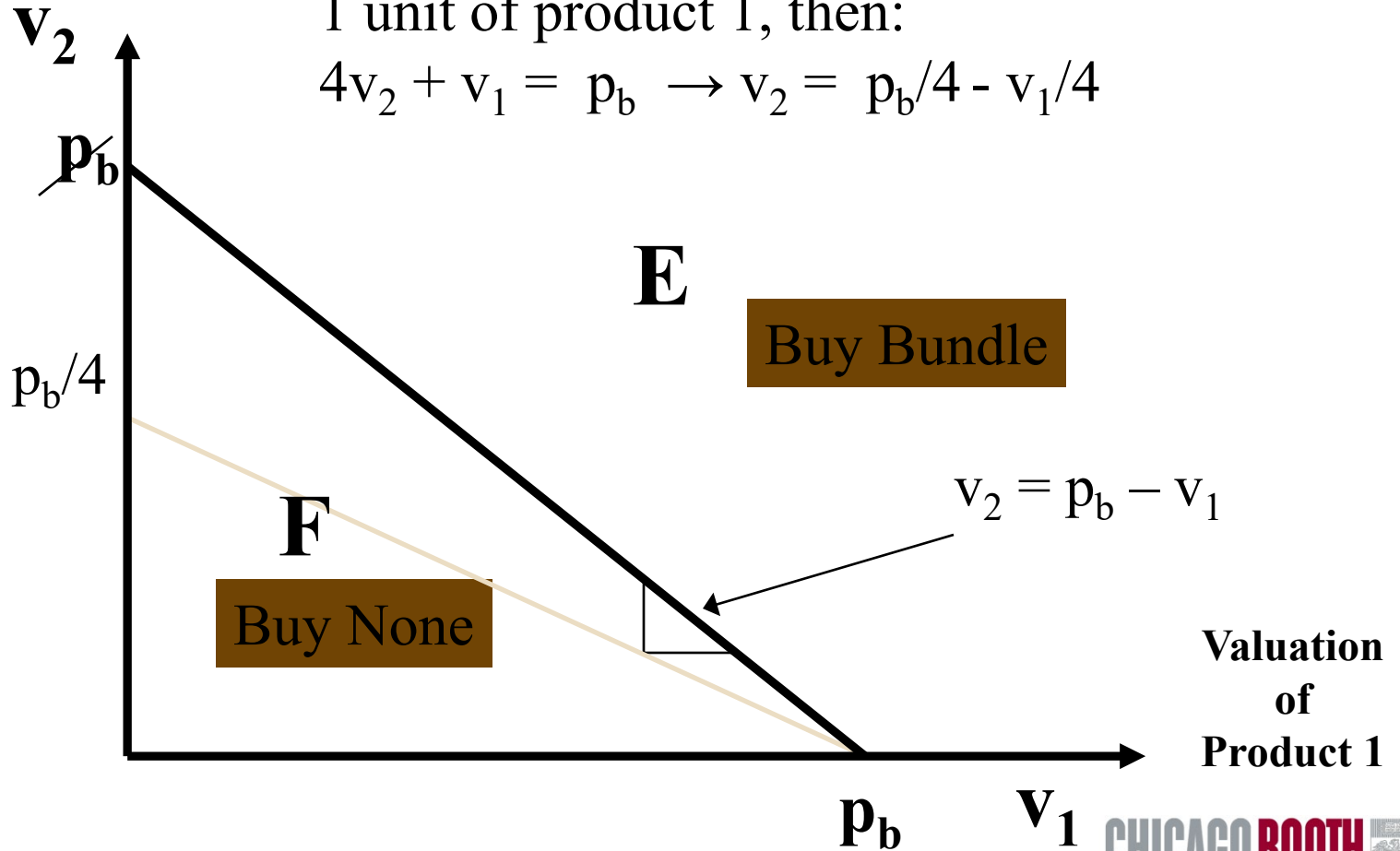
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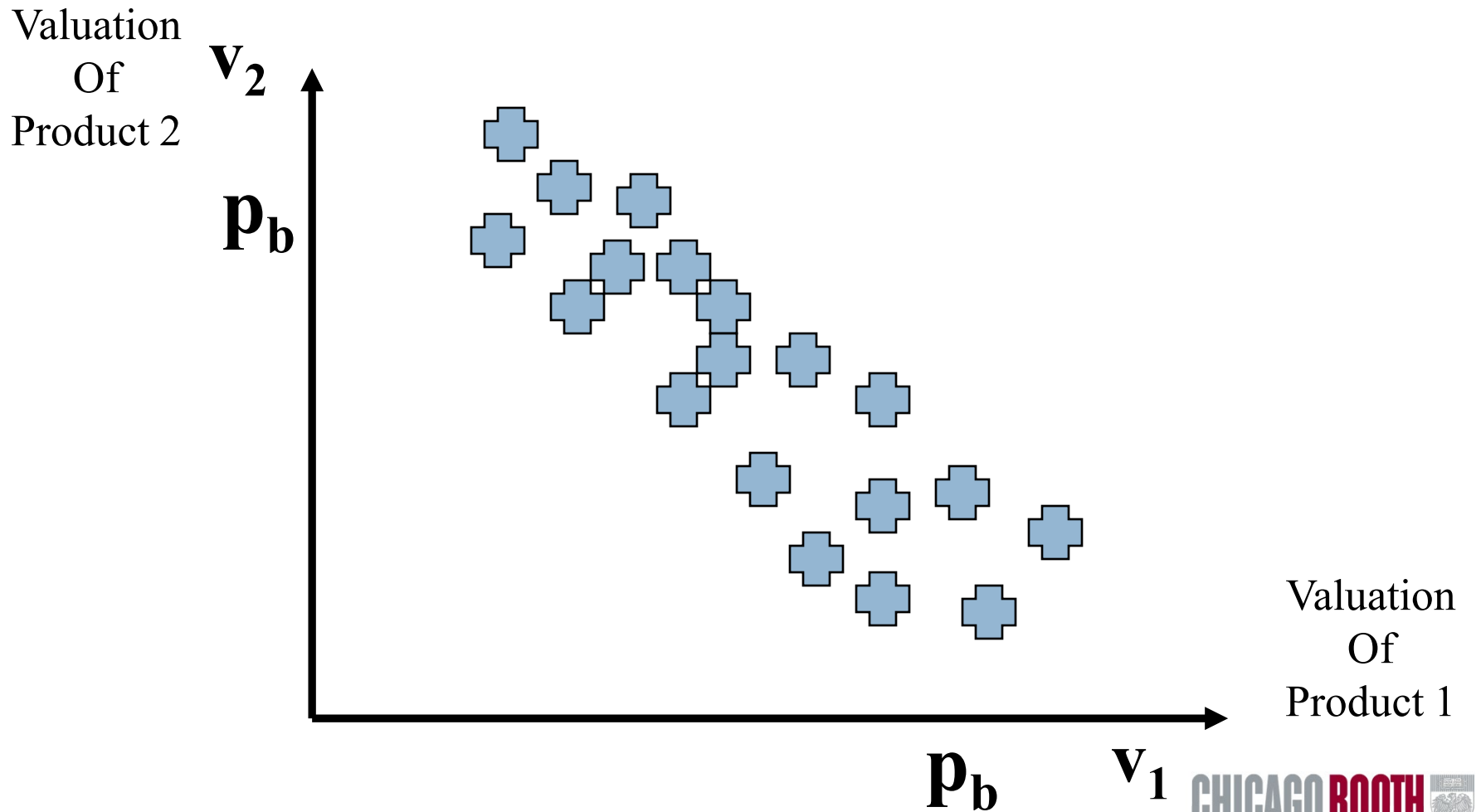
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Valuation
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Product 2



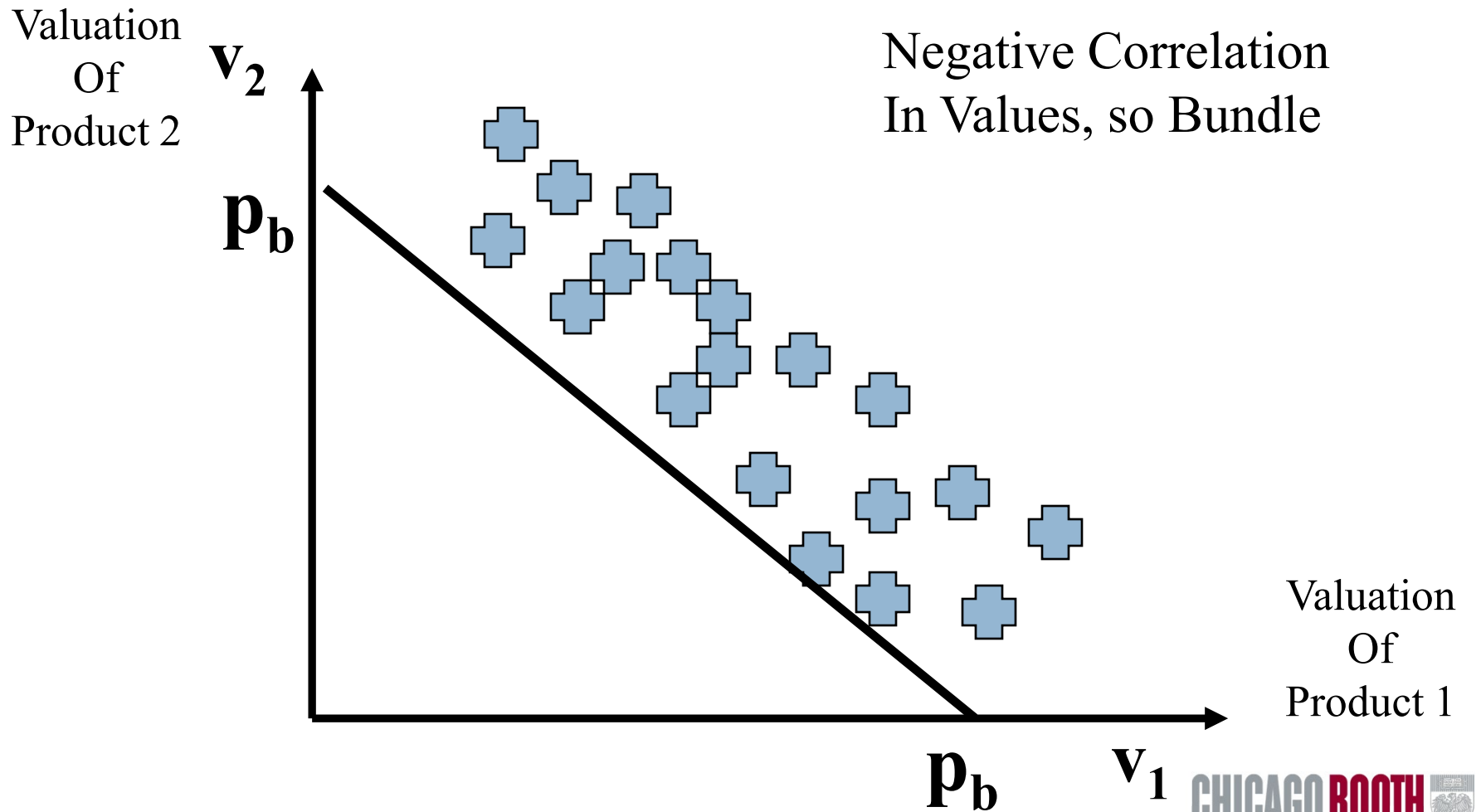
Bundle or Items?

44



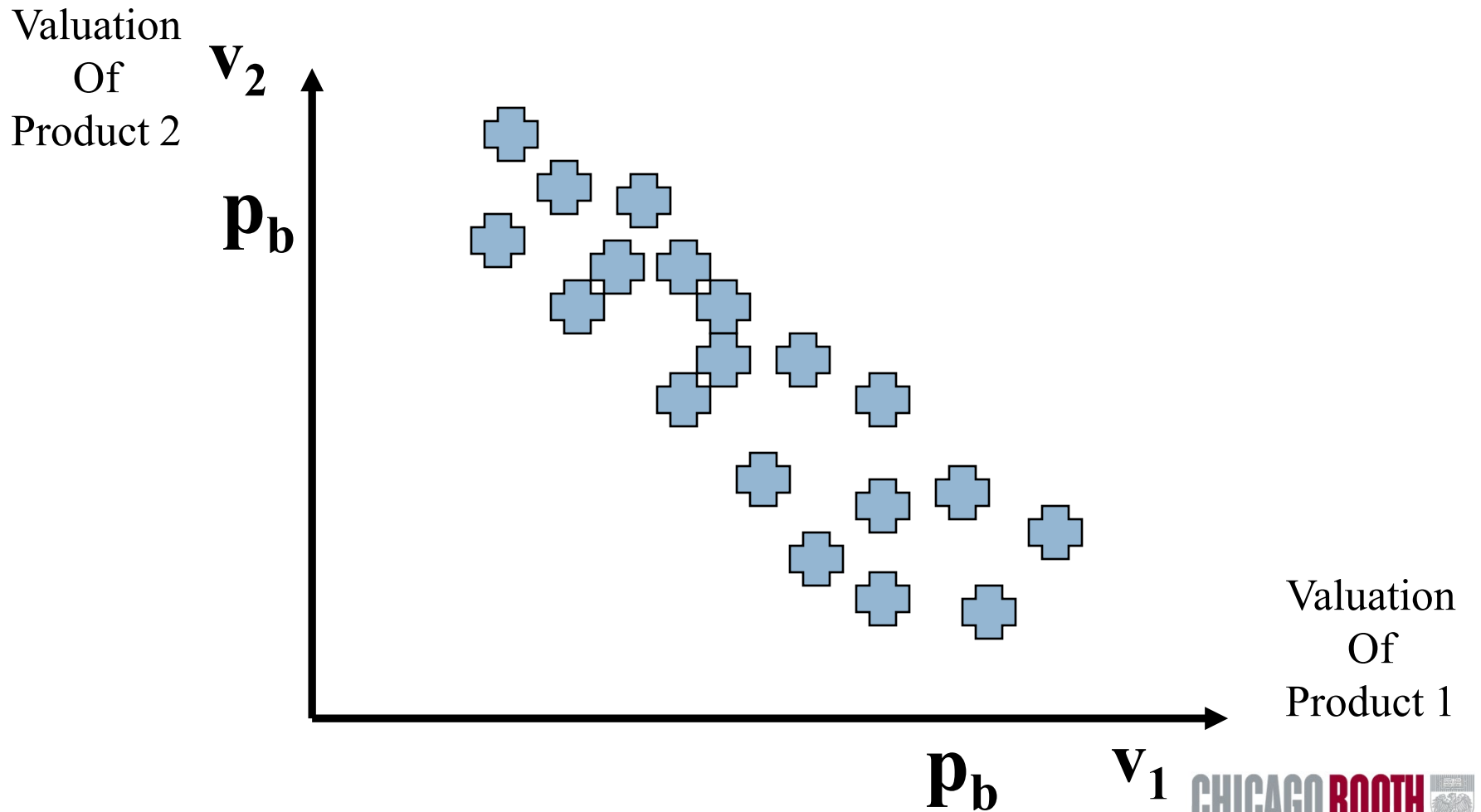
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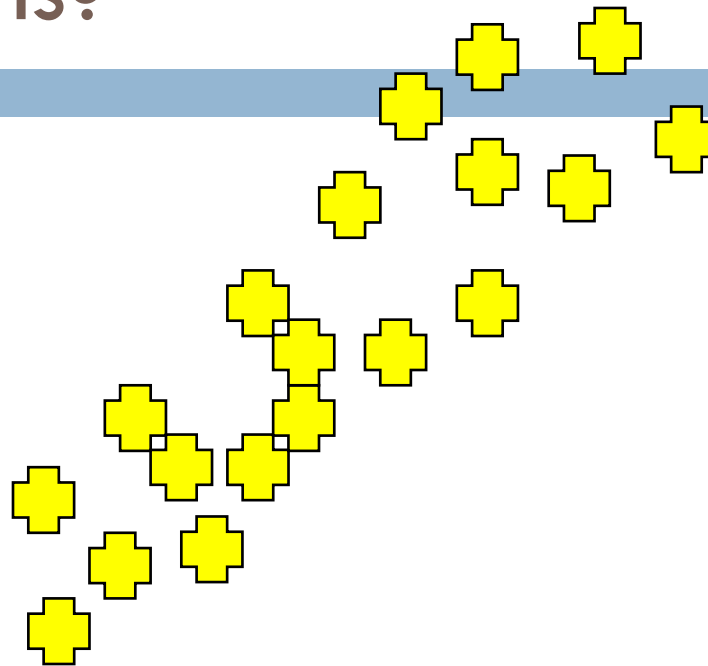


Bundle or Items?

45

Valuation
Of
Product 2

v_2

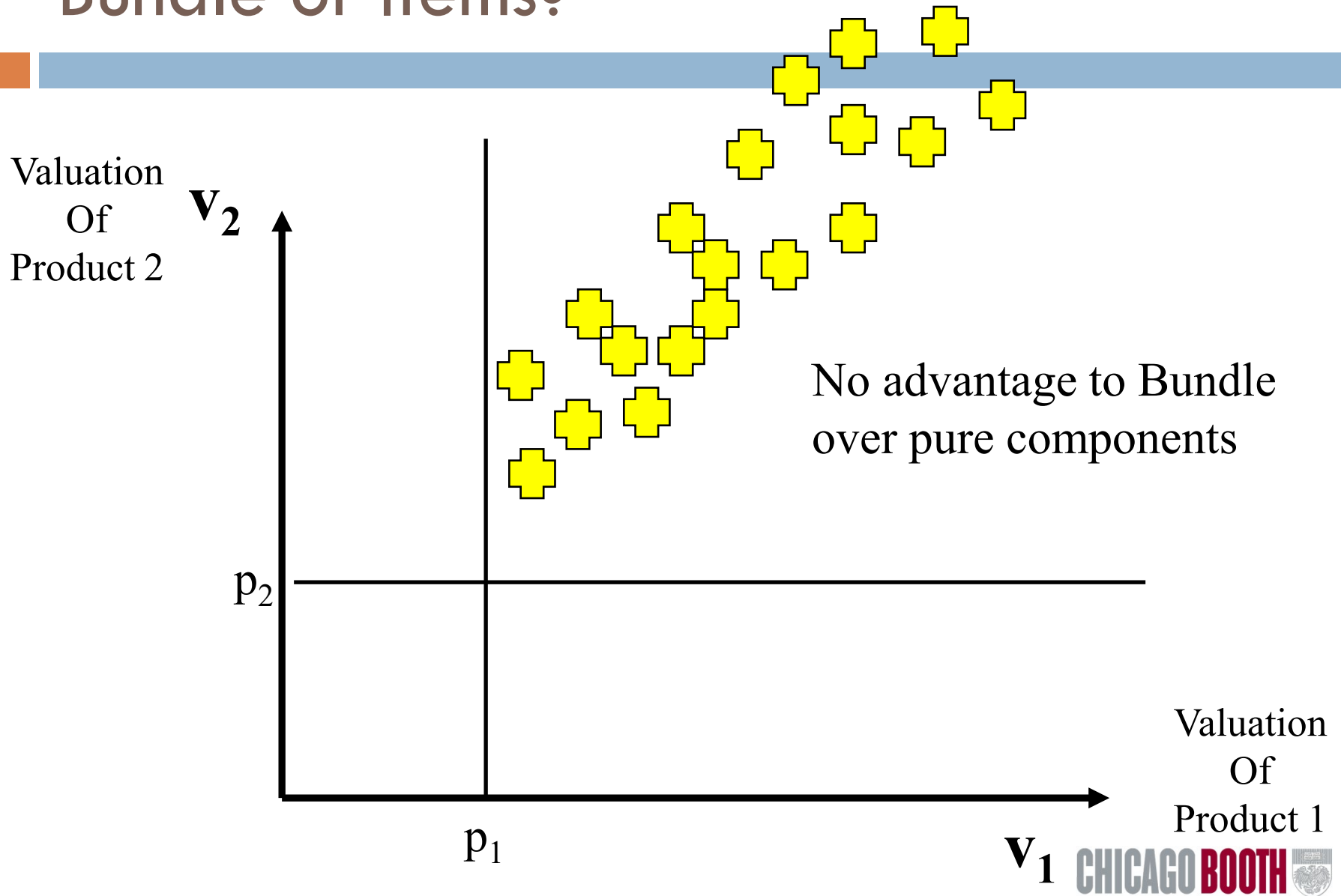


Valuation
Of
Product 1

v_1

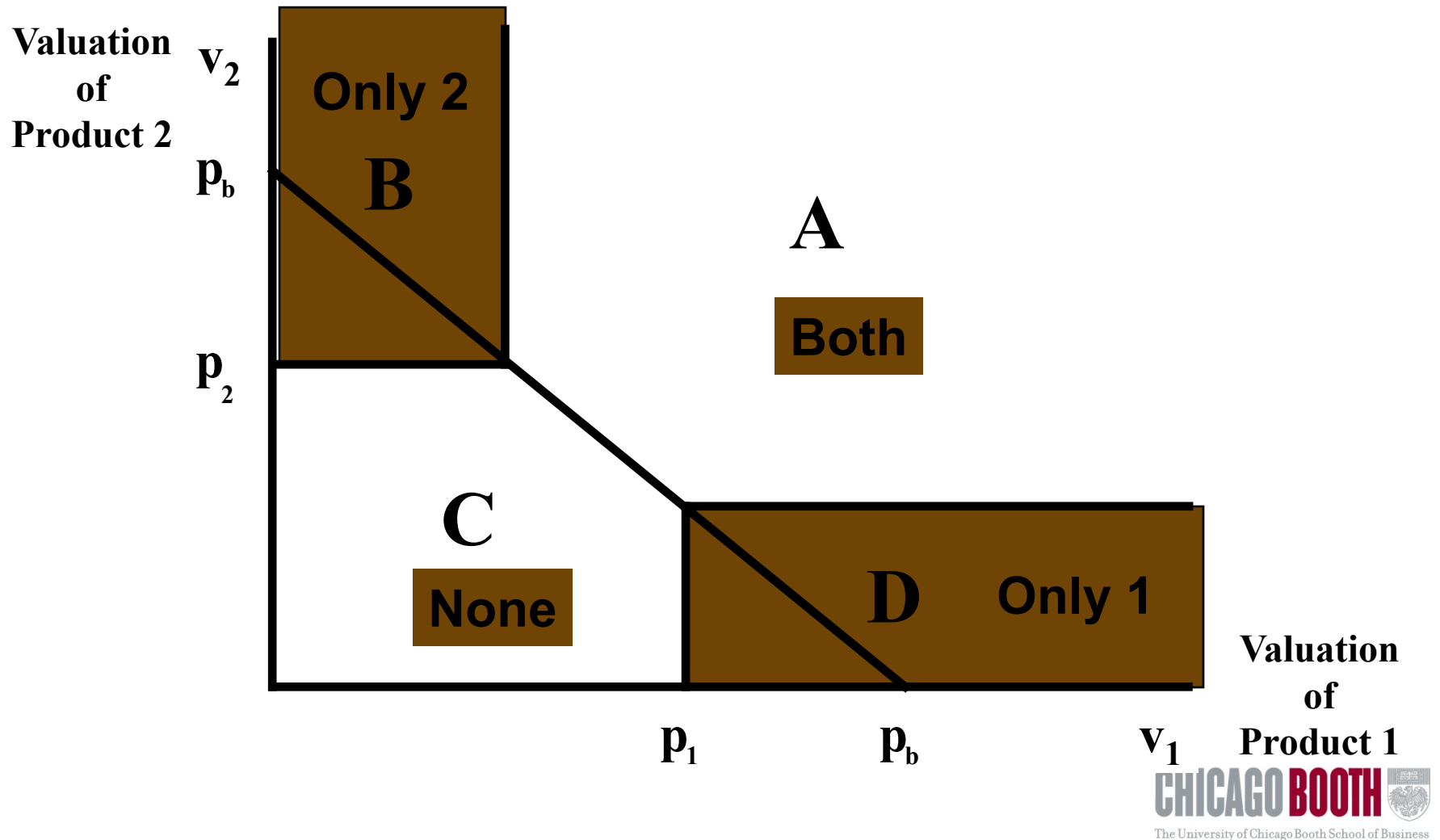
Bundle or Items?

45



Mixed Bundling Strategy

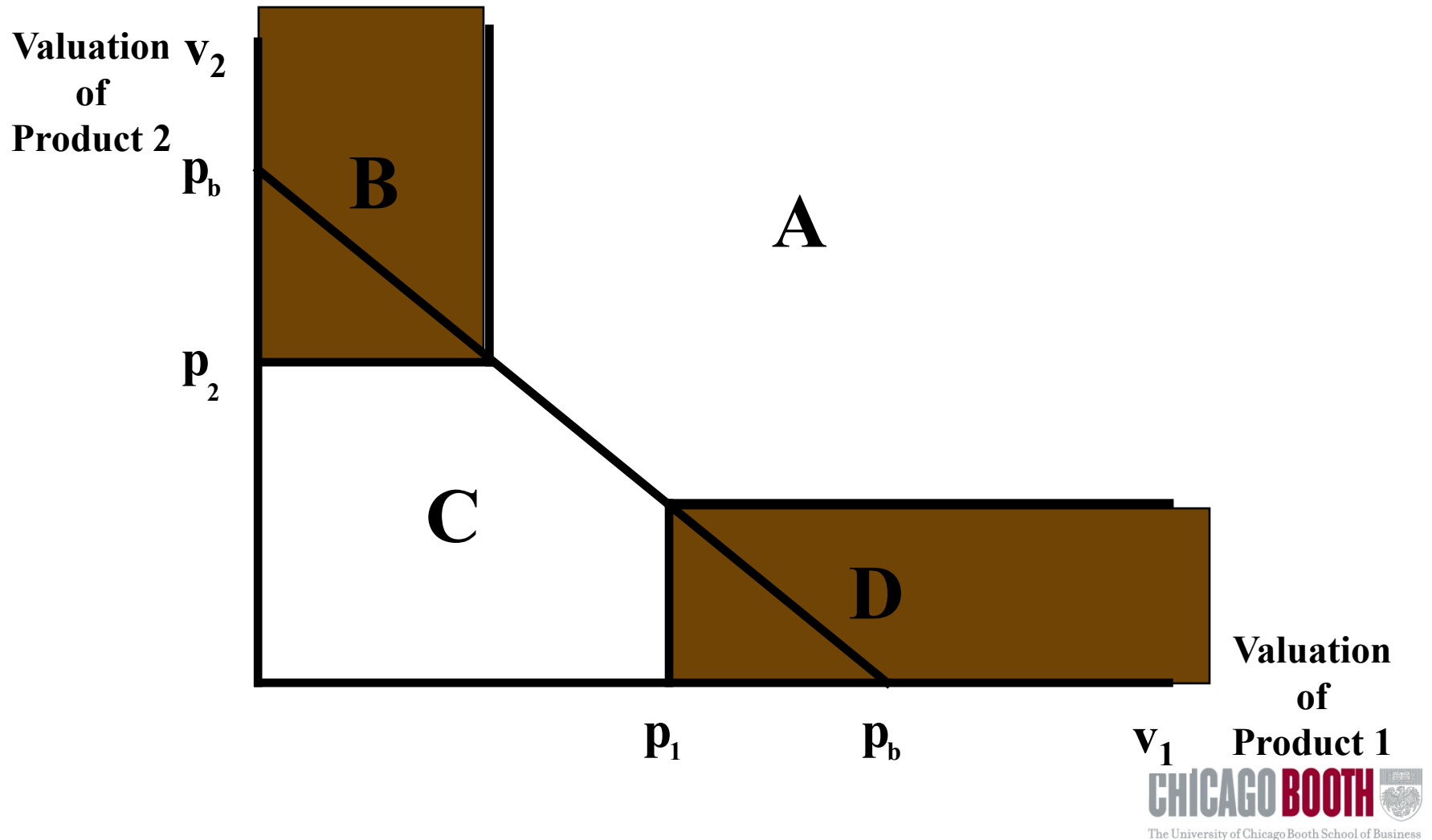
46



Mixed Bundling Tactics

Lower item 1 price from p_1 to p^*

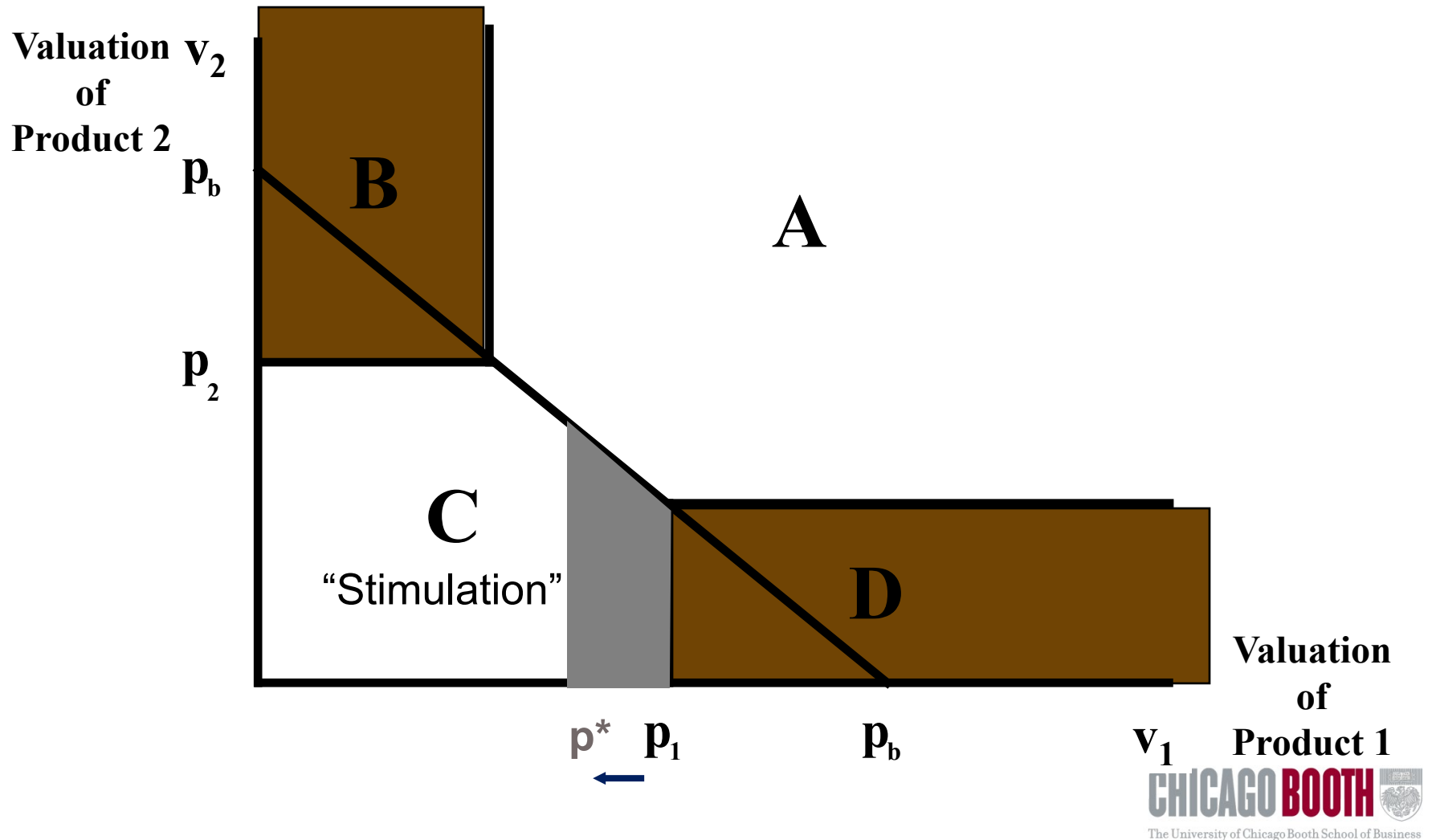
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